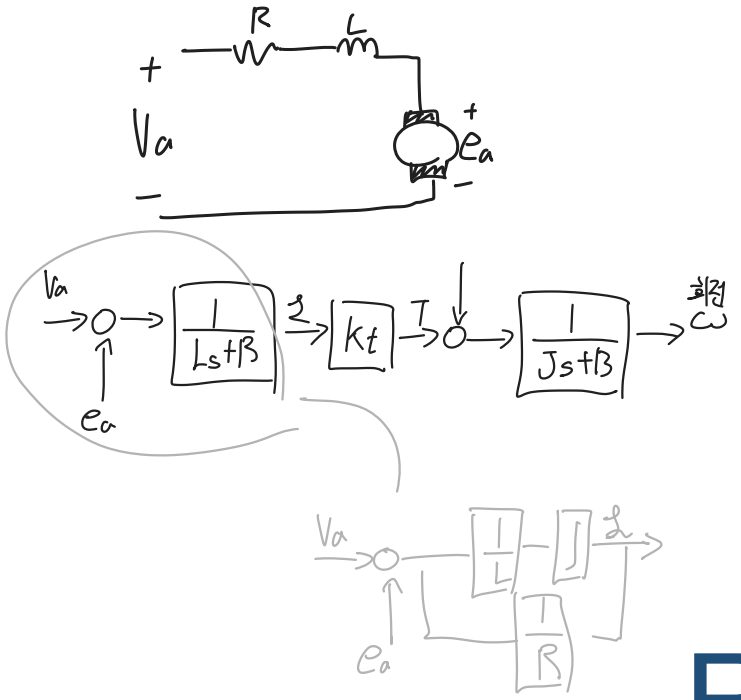


제어공학2
Automatic Control

Root-Locus Analysis

CHANGWON NATIONAL UNIVERSITY

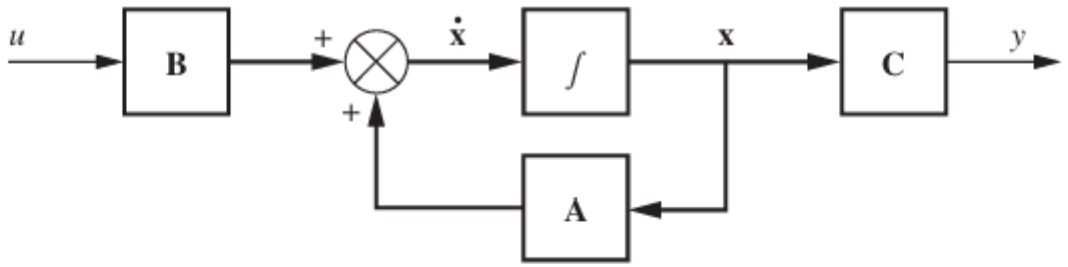
CHANGWON NATIONAL UNIVERSITY

상태피드백 제어

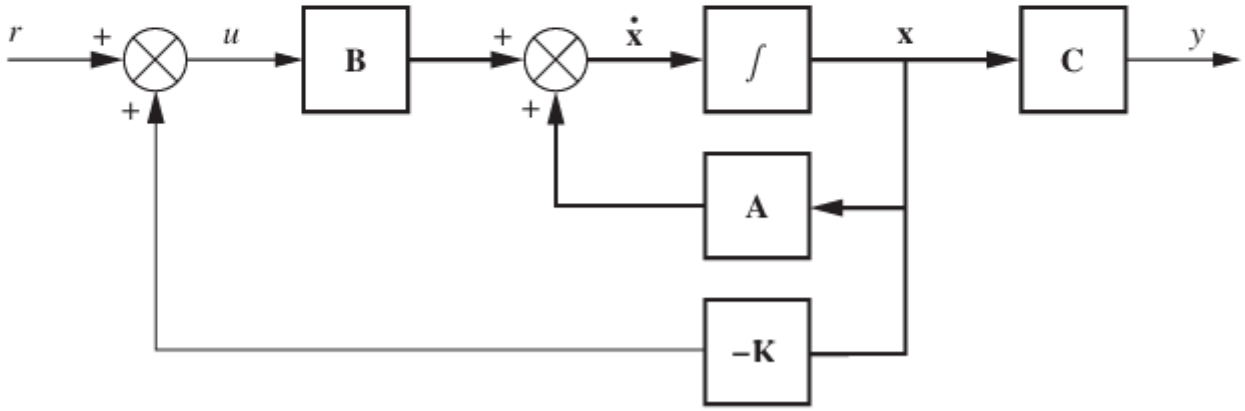
$$s^n + a_{n-1}s^{n-1} + \cdots + a_1 s + a_0 = 0$$

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u$$

$$y = \mathbf{C}\mathbf{x}$$



(a)



(b)

$$\frac{d\mathbf{x}(t)}{dt} = (\mathbf{A} - \mathbf{B}\mathbf{K})\mathbf{x}(t) + \mathbf{B}r(t)$$

CHANGWON NATIONAL UNIVERSITY

상태피드백을 이용한 극배치 설계

$$\frac{d\mathbf{x}(t)}{dt} = (\mathbf{A}-\mathbf{BK})\mathbf{x}(t)+\mathbf{B}r(t)$$

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ -a_0 & -a_1 & -a_2 & \cdots & -a_{n-1} \end{bmatrix}; \; B = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix};$$

$$\mathbf{C} = [\, c_1 \quad c_2 \quad \cdots \quad c_n \,]$$

적절한 K 값으로 pole을..?

$$s^n + a_{n-1}s^{n-1} + \cdots + a_1 s + a_0 = 0$$

$$\det(s\mathbf{I} - (\mathbf{A} - \mathbf{BK})) = s^n + (a_{n-1} + k_n)s^{n-1} + (a_{n-2} + k_{n-1})s^{n-2} + \cdots (a_1 + k_2)s + (a_0 + k_1) = 0$$

$$s^n + d_{n-1}s^{n-1} + d_{n-2}s^{n-2} + \cdots + d_2 s^2 + d_1 s + d_0 = 0$$

$$\mathbf{A} - \mathbf{BK} = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ -(a_0 + k_1) & -(a_1 + k_2) & -(a_2 + k_3) & \cdots & -(a_{n-1} + k_n) \end{bmatrix}$$

$$d_i = a_i + k_{i+1} \quad i = 0,\ 1,\ 2, \ldots,\ n-1$$

$$k_{i+1} = d_i - a_i$$

$\frac{Y}{U} = \frac{20s+100}{s^3+5s^2+4s} \Rightarrow [s^3+5s^2+4s]Y = [20s+100]U$

$\Rightarrow \frac{1}{s^3}(s^3Y+5s^2Y+4sY) = (20sU+100U)\frac{1}{s^3}$

$\Rightarrow Y = -\frac{5}{s}Y - \frac{4}{s^2}Y + \frac{20}{s^2}U + \frac{100}{s^3}U$ ($=x_1$)

$= \frac{1}{s}[-5Y + \frac{1}{s}[-4Y + 20U + \frac{1}{s}100U]]$ ($=x_3$, $=x_2$)

$\Rightarrow \begin{cases} X_1 = -\frac{5}{s}Y + \frac{1}{s}X_2 \\ sX_1 = -5Y + X_2 \\ \therefore \dot{x}_1 = -5y + x_2 \end{cases} \begin{cases} X_2 = \frac{1}{s}[-4x_1 + 20U + x_3] \\ sX_2 = -4X_1 + 20U + X_3 \\ \therefore \dot{x}_2 = -4x_1 + 20u + x_3 \end{cases} \begin{cases} X_3 = \frac{1}{s}100U \\ sX_3 = 100U \\ \therefore \dot{x}_3 = 100U \end{cases}$

CHANGWON NATIONAL UNIVERSITY

상태피드백을 이용한 극배치 설계

✓Design the phase variable feedback gains to yield 9.5% overshoot and a settling time of 0.74 second

$s = -5.4 \pm j7.2 \; and \; -5.1$

$(s+5.4+j7.2)(s+5.4-j7.2)(s+5.1)$

$$\frac{Y}{U} = G(s) = \frac{20(s+5)}{s(s+1)(s+4)} = \frac{X}{U}\frac{Y}{X} = \frac{1}{s(s+1)(s+4)}\,20(s+5)$$

$$T(s) = \frac{20(s+5)}{s^3 + 15.9s^2 + 136.08s + 413.1}$$

① $\frac{X}{U} = \frac{1}{s(s+1)(s+4)}$

$[s^3+5s^2+4s]X = U$ ($x_2 = \dot{x}_1$, $x_3 = \dot{x}_2 = \ddot{x}_1$)

$\Rightarrow \dot{x}_3 + 5x_3 + 4x_2 = u$

$$\therefore \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -4 & -5 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}u$$

② $\frac{Y}{X} = 20s + 100$

$Y = 20sX + 100X$

$$Y = \begin{bmatrix} 100 & 20 & 0 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -k_1 & -(4+k_2) & -(5+k_3) \end{bmatrix}\mathbf{x} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} r$$

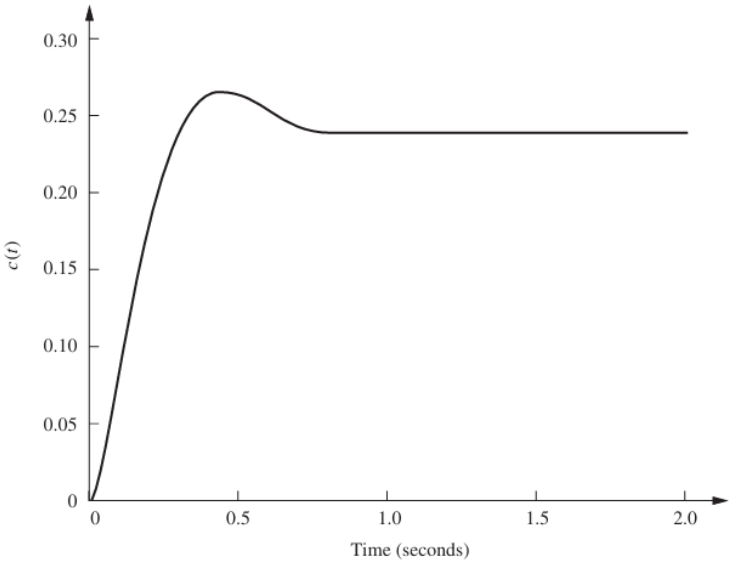
$$y = [\,100 \quad 20 \quad 0\,]\,\mathbf{x}$$

$$s^3 + 15.9s^2 + 136.08s + 413.1 = 0$$

$$\det(s\mathbf{I} - (\mathbf{A} - \mathbf{BK})) = s^3 + (5 + k_3)s^2 + (4 + k_2)s + k_1 = 0$$

($5 + k_3 = 15.9$, $4 + k_2 = 136.08$, $k_1 = 413.1$)

$$k_1 = 413.1; \quad k_2 = 132.08; \quad k_3 = 10.9$$



CHANGWON NATIONAL UNIVERSITY

관측기 설계

✓Closed-loop observer

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u$$

$$y = \mathbf{C}\mathbf{x}$$

$$\dot{\hat{\mathbf{x}}} = \mathbf{A}\hat{\mathbf{x}} + \mathbf{B}u$$

$$\hat{\mathbf{y}} = \mathbf{C}\hat{\mathbf{x}}$$

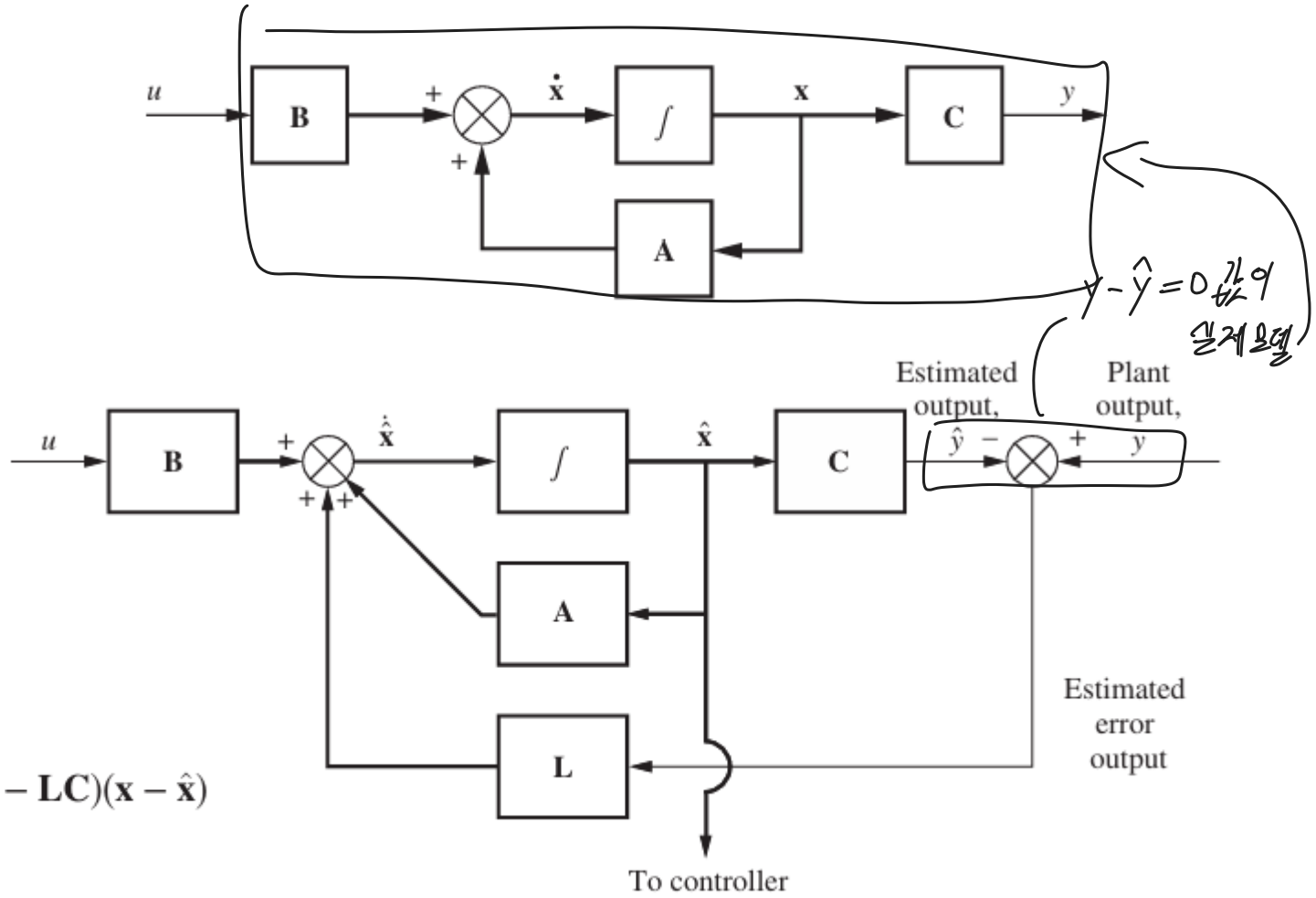
$$\dot{\mathbf{x}} - \dot{\hat{\mathbf{x}}} = \mathbf{A}(\mathbf{x} - \hat{\mathbf{x}})$$

$$y - \hat{y} = \mathbf{C}(\mathbf{x} - \hat{\mathbf{x}})$$

$$\dot{\hat{\mathbf{x}}} = \mathbf{A}\hat{\mathbf{x}} + \mathbf{B}u + \mathbf{L}(y - \hat{y})$$

$$\hat{y} = \mathbf{C}\hat{\mathbf{x}}$$

$$(\dot{\mathbf{x}} - \dot{\hat{\mathbf{x}}}) = (\mathbf{A} - \mathbf{LC})(\mathbf{x} - \hat{\mathbf{x}})$$



CHANGWON NATIONAL UNIVERSITY

관측기 설계 ∝F

✓Closed-loop observer

$$\dot{\hat{\mathbf{x}}} = \mathbf{A}\hat{\mathbf{x}} + \mathbf{B}u + \mathbf{L}(y - \hat{y})$$

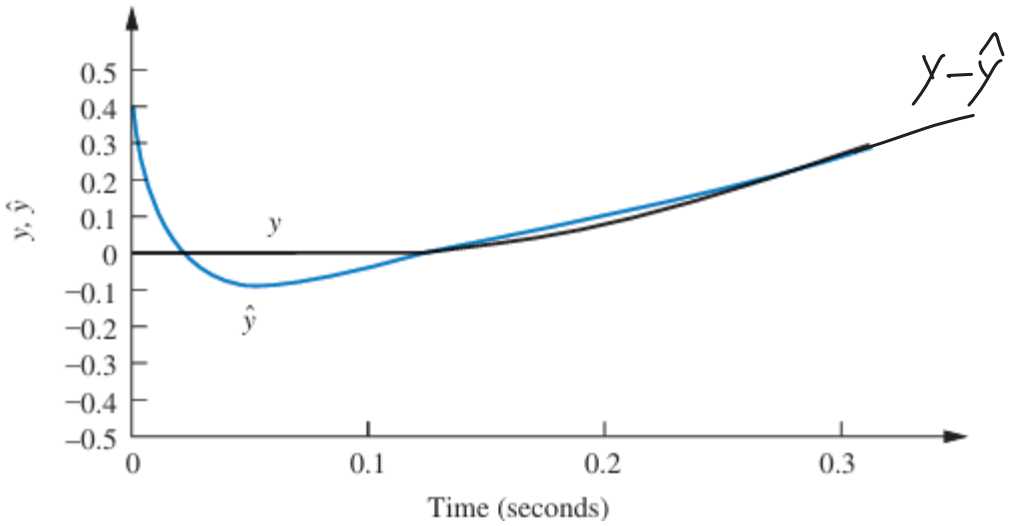
$$\hat{y} = \mathbf{C}\hat{\mathbf{x}}$$

$$\mathbf{A} - \mathbf{LC} = \begin{bmatrix} -a_{n-1} & 1 & 0 & 0 & \cdots & 0 \\ -a_{n-2} & 0 & 1 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ -a_1 & 0 & 0 & 0 & \cdots & 1 \\ -a_0 & 0 & 0 & 0 & \cdots & 0 \end{bmatrix} - \begin{bmatrix} l_1 \\ l_2 \\ \vdots \\ l_{n-1} \\ l_n \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 & \cdots & 0 \end{bmatrix}$$

$$= \begin{bmatrix} -(a_{n-1} + l_1) & 1 & 0 & 0 & \cdots & 0 \\ -(a_{n-2} + l_2) & 0 & 1 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ -(a_1 + l_{n-1}) & 0 & 0 & 0 & \cdots & 1 \\ -(a_0 + l_n) & 0 & 0 & 0 & \cdots & 0 \end{bmatrix}$$

$$s^n + (a_{n-1} + l_1)s^{n-1} + (a_{n-2} + l_2)s^{n-2} + \cdots + (a_1 + l_{n-1})s + (a_0 + l_n) = 0$$

$$s^n + a_{n-1}s^{n-1} + a_{n-2}s^{n-2} + \cdots + a_1 s + a_0 = 0$$



CHANGWON NATIONAL UNIVERSITY

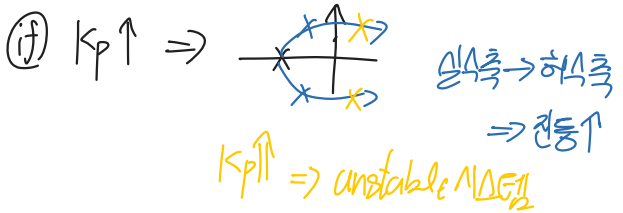
근궤적 기법

- ✓시스템의 폐루프 극에서 변수 변화의 효과를 결정하기 위한 근궤적을 공식화하고 해석
- ✓근궤적의 성질을 근거로 하여 직접 구성
- ✓변수가 여러 개인 시스템의 근컨투어 구성
- ✓근궤적과 근컨투어를 구성하기 위해 MATLAB 이용
- ✓제어 시스템을 설계하기 위해 근궤적 혹은 근컨투어 이용



$s^2+2s+1=0$

$s^2+2s+1+Kp=0$



CHANGWON NATIONAL UNIVERSITY

근궤적 기법

- ✓폐루프 전달함수의 극인 특성방정식의 근들은 선형 시스템의 절대 및 상대안정도를 결정
- ✓폐루프 전달함수의 극과 영점이 시스템의 동적 성능에 영향
- ✓근궤적 Root loci : -∞<K< ∞의 전체 근궤적
- ✓근컨투어 Root controur : 두 개 이상의 파라미터가 변할 때의 근궤적

$$F(s) = P(s) + KQ(s) = 0$$

$$P(s) = s^n + a_{n-1}s^{n-1} + \cdots + a_1 s + a_0$$

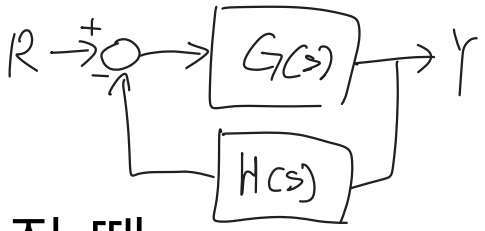
$$Q(s) = s^m + b_{m-1}s^{m-1} + \cdots + b_1 s + b_0$$

- K : 실정수, $-\infty < K < \infty$ → 나는 제어적으로 목의미 ⇒ $0 < |K| < \infty$
- n, m : 양의 정수
- $a_0, a_1, \ldots, a_{n-1}, b_0, b_1, \ldots, b_{m-1}$: 실수, 일정

CHANGWON NATIONAL UNIVERSITY

근궤적의 기본 성질

- ✓폐루프 전달함수
- ✓G(s)H(s)가 파라미터 K를 가질 때



$$\frac{Y(s)}{R(s)}=\frac{G(s)}{1+G(s)H(s)}$$

$$1+G(s)H(s)=0$$

$$G(s)H(s)=\frac{KQ(s)}{P(s)}$$

$$1+\frac{KQ(s)}{P(s)}=\frac{P(s)+KQ(s)}{P(s)}=0$$

$s^3+3s^2+2s+s^2+3s+2Ks+5=0$

$s^3+4s^2+5s+5+2Ks=0$

$1+\frac{K\cdot 2s}{s^3+4s^2+5s+5}=-1=0$

$\frac{2s}{s^3+4s^2+5s+5}=-\frac{1}{K}$

= G, H, 으로 두고 원하는 K 찾기

$$s(s+1)(s+2)+s^2+(3+2K)s+5=0$$

$$1+\frac{2Ks}{s(s+1)(s+2)+s^2+3s+5}=0$$

$$\frac{Q(s)}{P(s)}=\frac{2s}{s^3+4s^2+5s+5}$$

CHANGWON NATIONAL UNIVERSITY

근궤적의 기본 성질

- ✓폐루프 전달함수
- ✓G(s)H(s)를 다음과 같이 표기
- ✓$G_1(s)H_1(s)$은 K를 포함하지 않음

$$G(s)H(s) = KG_1(s)H_1(s)$$

$$1 + G(s)H(s) = 0$$

$$G_1(s)H_1(s) = -\frac{1}{K}$$

s : σ+jω 형태복소수

크기조건

$$|G_1(s)H_1(s)| = \frac{1}{|K|} \quad -\infty < K < \infty$$

각도조건

$$\angle G_1(s)H_1(s) = (2i+1)\pi \quad K \geq 0$$

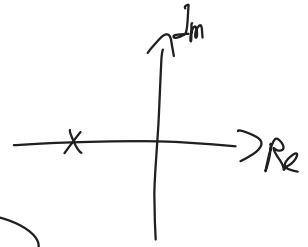
= odd multiples of π radians or 180°

$$\angle G_1(s)H_1(s) = 2i\pi \quad K \leq 0$$

= even multiples of π radians or 180°

where $i = 0, \pm 1, \pm 2, \ldots$ (any integer).

180°, 540° …



CHANGWON NATIONAL UNIVERSITY

근궤적의 기본 성질

✓근궤적의 도해적 구성 Graphical construction of the Root-locus

$$G(s)H(s) = KG_1(s)H_1(s) = \frac{K(s+z_1)(s+z_2)\cdots(s+z_m)}{(s+p_1)(s+p_2)\cdots(s+p_n)}$$

$$|G_1(s)H_1(s)| = \frac{\prod_{i=1}^{m}|s+z_i|}{\prod_{k=1}^{n}|s+p_k|} = \frac{1}{|K|} \quad -\infty < K < \infty$$

For $0 \le K < \infty$:

$$\angle G_1(s)H_1(s) = \sum_{k=1}^{m}\angle(s+z_k) - \sum_{j=1}^{n}\angle(s+p_j) = (2i+1)\times 180^\circ$$

For $-\infty < K \le 0$:

$$\angle G_1(s)H_1(s) = \sum_{k=1}^{m}\angle(s+z_k) - \sum_{j=1}^{n}\angle(s+p_j) = 2i\times 180^\circ$$

where $i = 0, \pm1, \pm2, \ldots$

$$Z_1 = r_1 e^{j\theta_1} \qquad Z_2 = r_2 e^{j\theta_2}$$

$$Z_1 \cdot Z_2 = r_1 \cdot r_2 \cdot e^{j(\theta_1+\theta_2)}$$

$$\frac{Z_1}{Z_2} = \frac{r_1}{r_2} e^{j(\theta_1-\theta_2)}$$

$$|K| = \frac{\prod_{j=1}^{n}|s+p_j|}{\prod_{i=1}^{m}|s+z_i|}$$

pole

zero

CHANGWON NATIONAL UNIVERSITY

근궤적의 기본 성질

✓근궤적의 도해적 구성 Graphical construction of the Root-locus

- S평면에서 모든 s_1이 양의 K에 대한 식을 만족시키는지 조사

$$G(s)H(s)=\frac{K(s+z_1)}{s(s+p_2)(s+p_3)}$$

: zero의 각은 분자에있고 → 모두더함

: pole의 각은 분모에 있으므로 → 부호빼줌

$$\angle(s_1+z_1)-\angle s_1-\angle(s_1+p_2)-\angle(s_1+p_3)=\theta_{z1}-\theta_{p1}-\theta_{p2}-\theta_{p3}=(2i+1)\times 180^\circ$$

조건(1)

180°홀수배

$s=0, -p_2, -p_3$

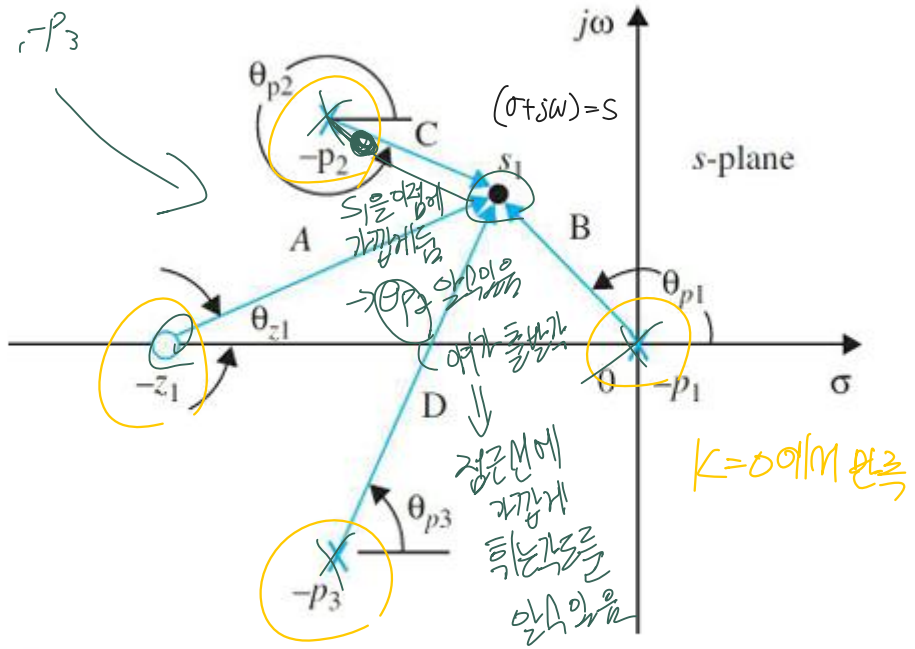


Figure 9-1 Pole-zero configuration of $G(s)H(s)=K(s+z_1)/[s(s+p_2)\times(s+p_3)]$.

- 근궤적 상 K의 크기

$$|K|=\frac{\prod_{j=1}^{n}|s+p_j|}{\prod_{i=1}^{m}|s+z_i|}$$

조건(2)

$$|K|=\frac{|s_1||s_1+p_2||s_1+p_3|}{|s_1+z_1|}=\frac{BCD}{A}$$

Pole 거리모두곱

Zero 거리모두곱

CHANGWON NATIONAL UNIVERSITY

근궤적의 성질

✓K=0과 K=±∞인 점

- 근궤적상 K=0인 점은 G(s)H(s)의 극점
- 근궤적상 K=±∞인 점은 G(s)H(s)의 영점
- 극과 영점은 필요하다면 무한대에 있는 점들도 포함

$$G_1(s)H_1(s) = -\frac{1}{K}$$

$$s(s+2)(s+3)+K(s+1)=0$$

$$1+G(s)H(s)=1+\frac{K(s+1)}{s(s+2)(s+3)}=0$$

$$G(s)H(s)=\frac{K(s+1)}{s(s+2)(s+3)}$$

① pole = 0, -2, -3 | K=0

② K = ∞

$$\frac{s(s+1)(s+3)}{\infty} + (s+1) = 0$$

pole = -1 | K=∞

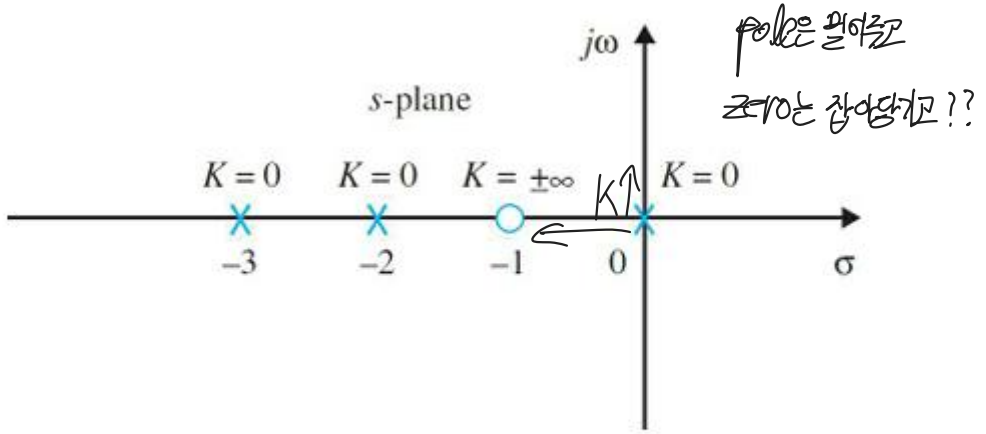


Figure 9-2 Points at which $K = 0$ and $K = \pm\infty$ on the *RL of* $s(s+2)(s+3)+K(s+1)=0$.

근궤적의 성질

✓근궤적의 지로의 수 Number of branches on the root loci

- 근궤적의 지로수는 다항식의 차수와 같음

$$F(s) = P(s) + KQ(s) = 0$$

- Ex) 식의 차수가 3이기 때문에 지로의 수는 3개

$$s(s+2)(s+3) + K(s+1) = 0$$

CHANGWON NATIONAL UNIVERSITY

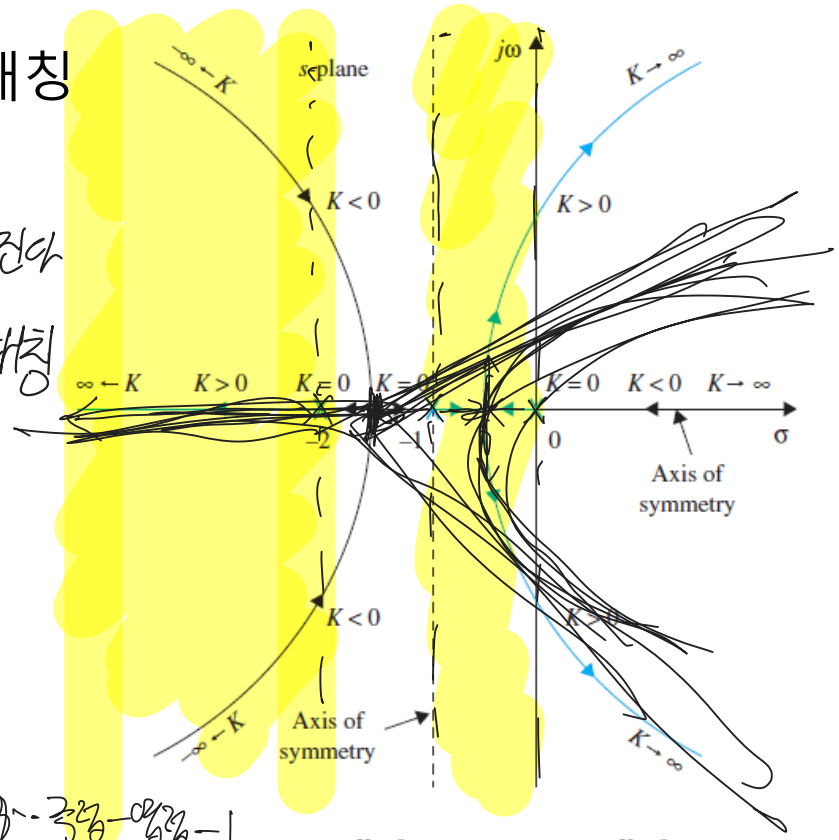
근궤적의 성질

✓근궤적의 지로의 수 Symmetry of the root loci

- 근궤적은 s평면의 실수축에 대하여 대칭
- G(s)H(s)의 극과 영점의 대칭축에 대하여 대칭

$$s(s+1)(s+2)+K=0$$

$$G(s)H(s)=\frac{K}{s(s+1)(s+2)}$$



CHANGWON NATIONAL UNIVERSITY

근궤적의 성질

✓근궤적의 지로의 수 Symmetry of the root loci

- 근궤적은 s평면의 실수축에 대하여 대칭
- G(s)H(s)의 극과 영점의 대칭축에 대하여 대칭

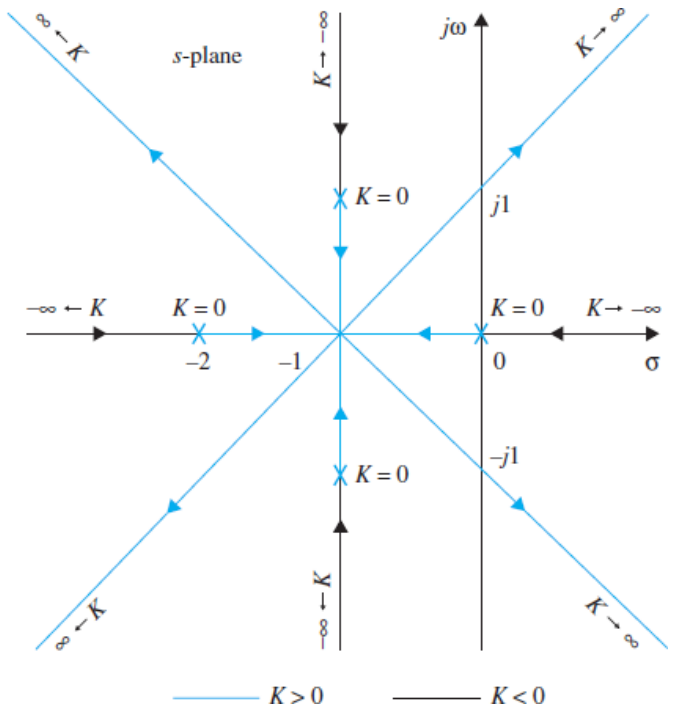
$$s(s+2)(s^2+2s+2)+K=0$$

$$1+\frac{K}{s(s+1)(s^2+2s+1)}=0$$

pole = 0, -2, -1±j

∴ 근이 4개 움직임...?

K 증가에 따라



CHANGWON NATIONAL UNIVERSITY

근궤적의 성질

✓근궤적의 점근선 각도: |s|=∞에서 근궤적의 성질

- S의 값이 커지면 K≥0에 대한 근궤적은 다음 식으로 주어지는 각도를 가진 점근선에 점근
- n과 m은 각각 G(s)H(s)의 유한 극점과 유한 영점의 수

$$\theta_i = \frac{(2i+1)}{|n-m|}\times 180^\circ \quad n \neq m \qquad \text{where } i=0,1,2,\ldots,|n-m|-1.$$

pole 수 n, zero 수 m

if $z=1$ $p=3$

$$\frac{(2i+1)}{3-1}180^\circ = 90^\circ \times (2i+1)$$

∴ 90° & 270° 각을으로 움직 (점근선)

✓점근선의 교차점(도심) Intersect of the asymptote

$$\sigma_1 = \frac{\sum \text{finite poles of } G(s)H(s) - \sum \text{finite zeros of } G(s)H(s)}{n-m}$$

$$\sigma_1 = \frac{\sum \text{real parts of poles of } G(s)H(s) - \sum \text{real parts of zeros of } G(s)H(s)}{n-m}$$

$$\frac{(s+z_n)(\cdots)}{(s+p_n)(\cdots)} = \frac{Ks^m}{s^n} \text{ 근사} \Rightarrow \frac{1}{s^{(n-m)}}$$

$$|n-m|\theta = (2i+1)180^\circ$$

$$\theta = \frac{(2i+1)180^\circ}{|n-m|}$$

CHANGWON NATIONAL UNIVERSITY

근궤적의 성질

✓근궤적의 점근선 각도: |s|=∞에서 근궤적의 성질

- S의 값이 커지면 K≥0에 대한 근궤적은 다음 식으로 주어지는 각도를 가진 점근선에 점근
- n과 m은 각각 G(s)H(s)의 유한 극점과 유한 영점의 수

$$s(s+2)(s+3)+K(s+1)=0$$

$$1+G(s)H(s)=1+\frac{K(s+1)}{s(s+2)(s+3)}=0$$

zeros 수 = 1

pole 수 = 3

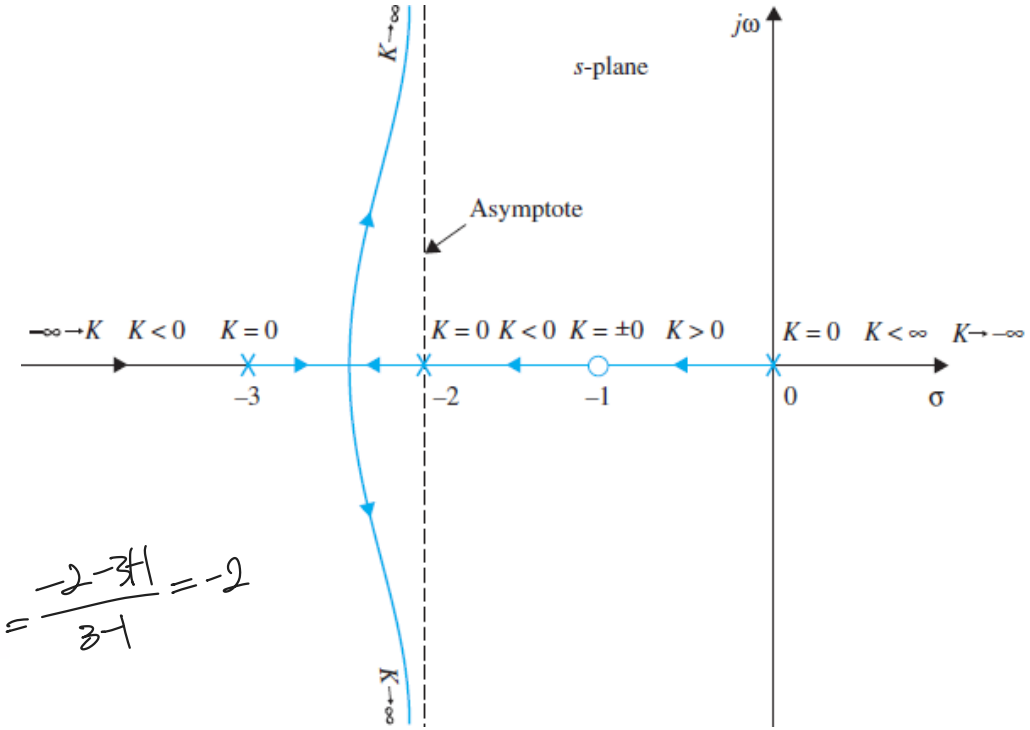
$$G(s)H(s)=\frac{K(s+1)}{s(s+2)(s+3)}$$

$$\theta_i=\frac{(2i+1)}{|n-m|}\times 180^\circ \quad n\neq m \qquad \text{where } i=0,1,2,\ldots,|n-m|-1.$$

$\frac{1}{3-1}\times 180^\circ = 90^\circ$ 각만 270°

$$\sigma_1=\frac{\sum \text{finite poles of } G(s)H(s)-\sum \text{finite zeros of } G(s)H(s)}{n-m}$$

$= \frac{-2-1}{3-1} = -2$



CHANGWON NATIONAL UNIVERSITY

근궤적의 성질

- ✓K=0: 근궤적에서 K=0에 대응하는 점들은 G(s)H(s)의 극점
- ✓K=±∞: 근궤적에서 K=±∞인 점들은 G(s)H(s)의 영점
- ✓4개의 근궤적 지로가 존재
- ✓근궤적은 실수축에 대칭
- ✓유한극의 수는 유한영점의 수를 3만큼 초과 → 3개의 근궤적이 s= ∞에 접근

$$s(s+4)(s^2+2s+2)+K(s+1)=0$$

$$G(s)H(s)=\frac{K(s+1)}{s(s+4)(s^2+2s+2)}$$

$$j=0:\quad \theta_0=\frac{180^\circ}{3}=60^\circ$$

$$j=1:\quad \theta_1=\frac{540^\circ}{3}=180^\circ$$

$$j=2:\quad \theta_2=\frac{900^\circ}{3}=300^\circ$$

CHANGWON NATIONAL UNIVERSITY

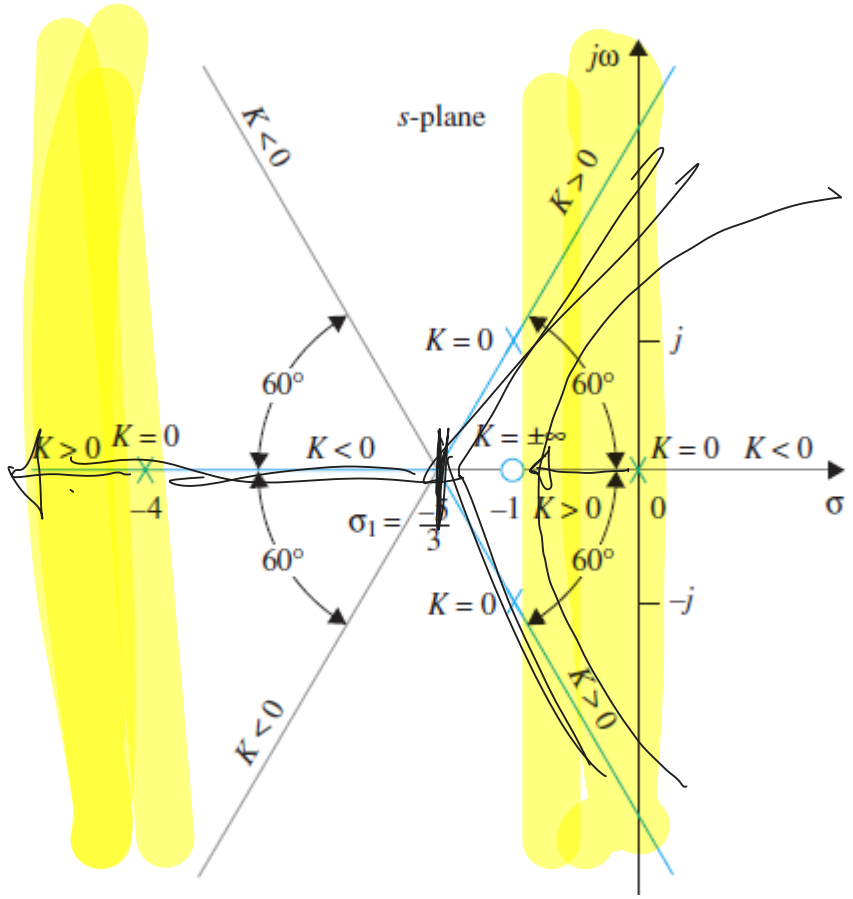
근궤적의 성질

✓교차점

$$s(s+4)(s^2+2s+2)+K(s+1)=0$$

$$G(s)H(s)=\frac{K(s+1)}{s(s+4)(s^2+2s+2)}$$

$$\sigma_1=\frac{(-4-1-1)-(-1)}{4-1}=-\frac{5}{3}$$



CHANGWON NATIONAL UNIVERSITY

근궤적의 성질

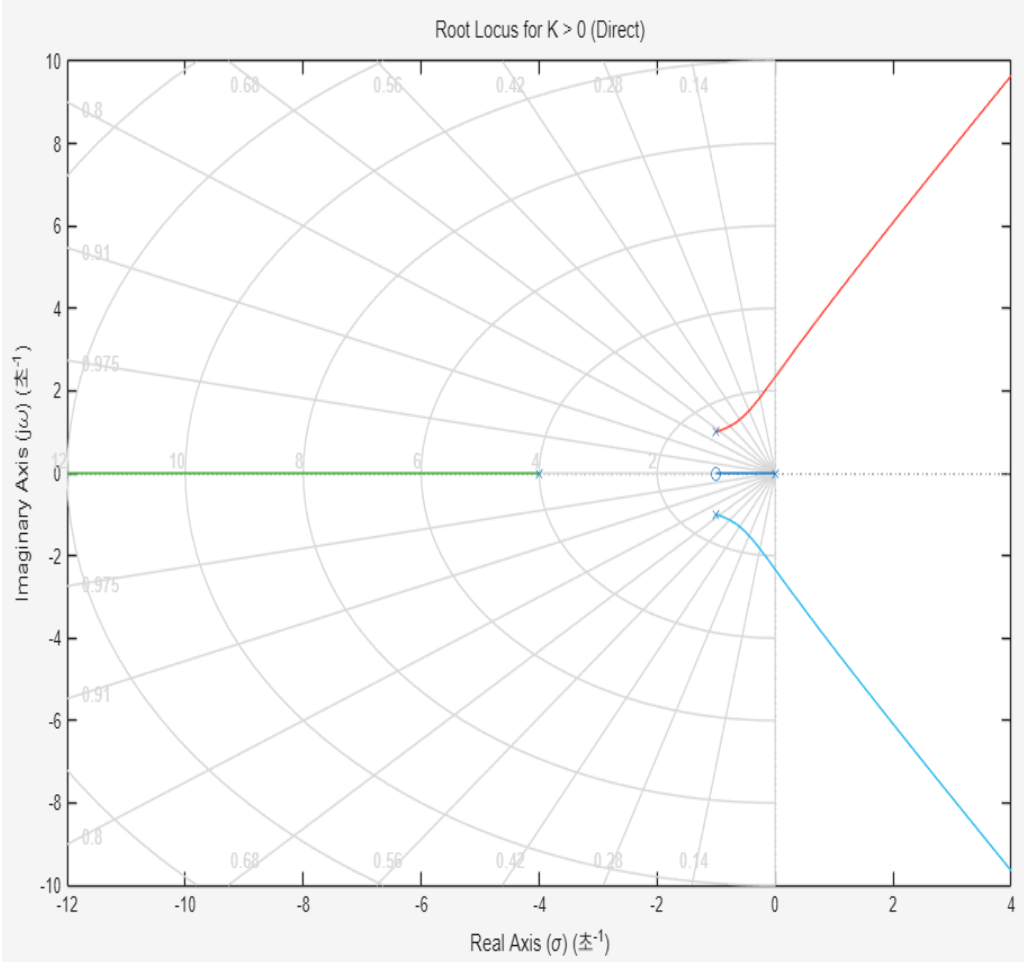
✓MATLAB

$$G(s)H(s)=\frac{K(s+1)}{s(s+4)(s^2+2s+2)}$$

```
% 라플라스 변수 s 정의
s = tf('s');

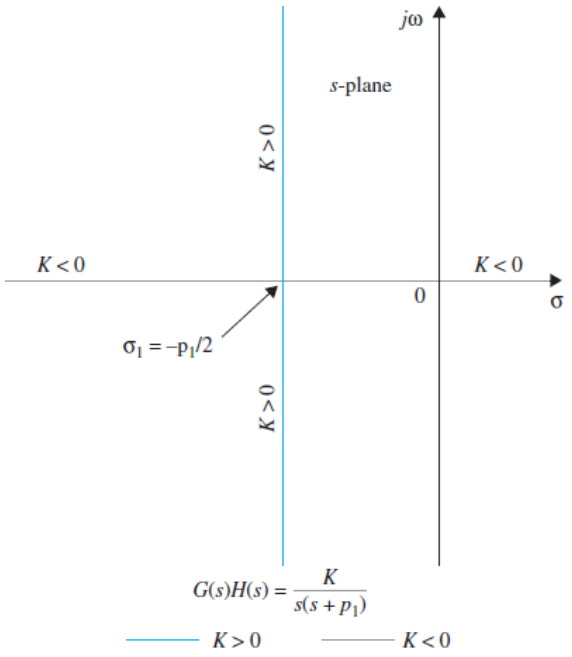
% 개루프 전달함수 G(s)H(s) 정의
G = (s + 1) / (s * (s + 4) * (s^2 + 2*s + 2));

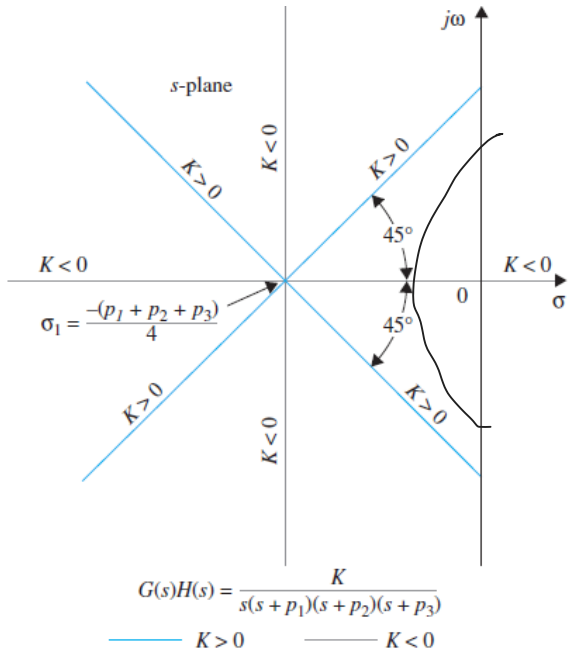
% 1. 양의 이득 (K > 0) 근궤적
figure;
rlocus(G);
grid on;
title('Root Locus for K > 0 (Direct)');
xlabel('Real Axis (\sigma)');
ylabel('Imaginary Axis (j\omega)');
```

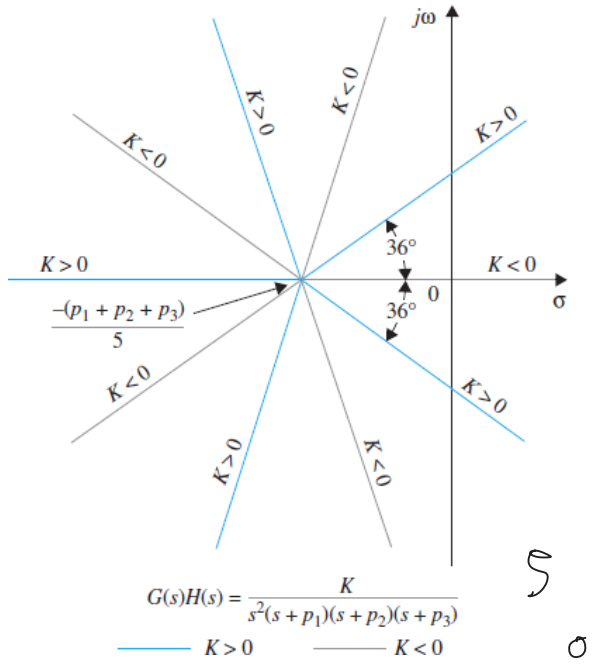


근궤적의 성질

✓근궤적의 점근선 예

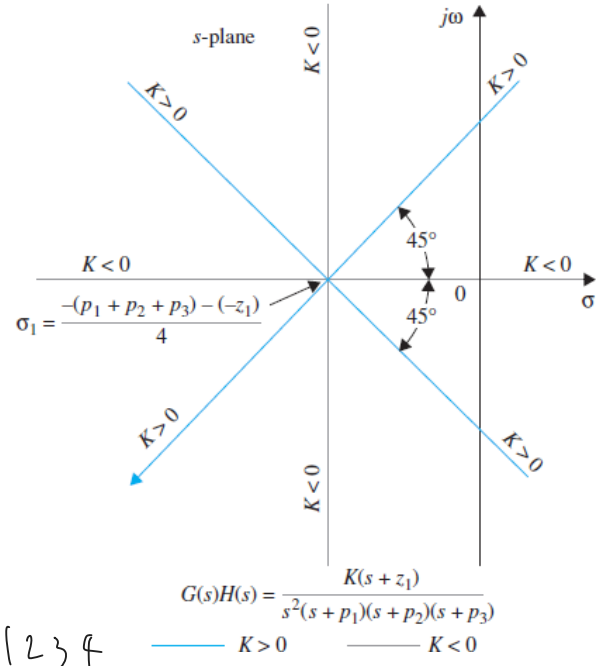






5

0 1 2 3 4



$z_b - o_{z_b} = 5$

CHANGWON NATIONAL UNIVERSITY

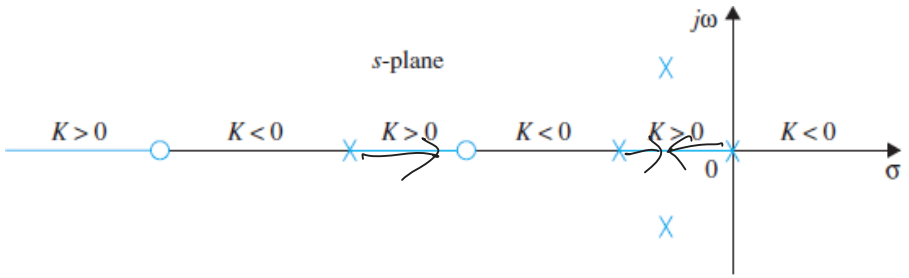
근궤적의 성질

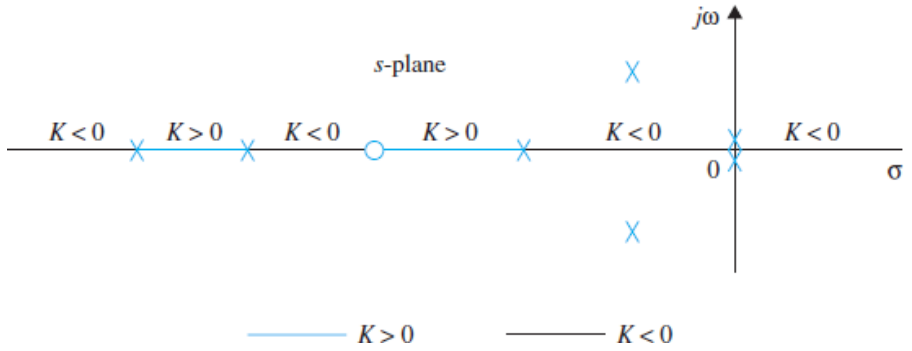
✓실수축 상의 근궤적

- 오직 G(s)H(s)의 실수값을 가지는 영점과 극만이 각에 대한 관계식에 영향을 줌
- s_1의 오른쪽에 놓여있는 오직 G(s)H(s)의 실수값을 가지는 영점과 극만이 각에 대한 관계식에 영향을 줌
- 근궤적 위에 있는 점 s_1에 대해 그 점의 오른쪽으로 G(s)H(s)의 극과 영점이 홀수개 존재

For $0 \le K < \infty$:

$$\angle G_1(s)H_1(s) = \sum_{k=1}^{m} \angle(s+z_k) - \sum_{j=1}^{n} \angle(s+p_j) = (2i+1)\times 180^\circ$$





CHANGWON NATIONAL UNIVERSITY

근궤적의 성질

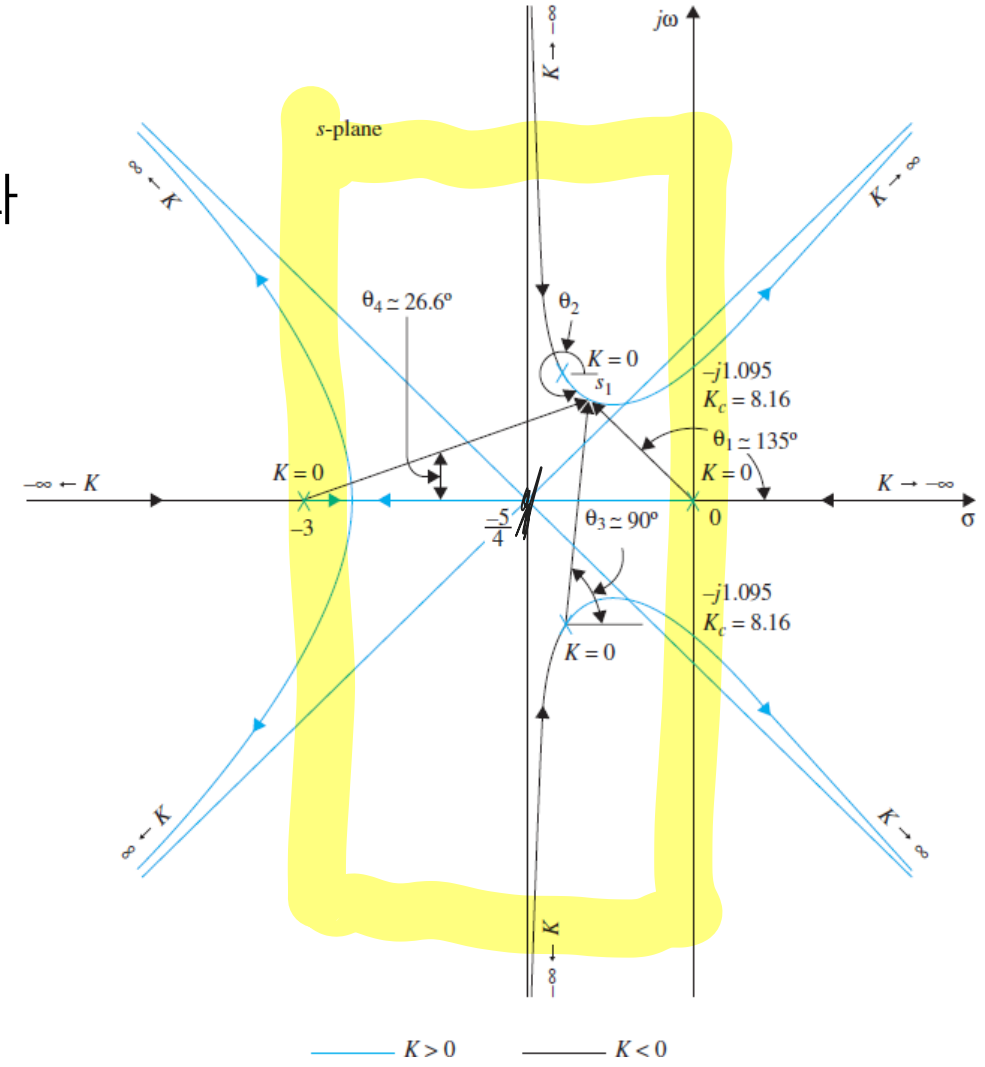
✓근궤적의 출발각과 도착각

- G(s)H(s)의 극이나 영점에서 근궤적의 출발각과 도착각은 각각 그 점 부근의 궤적에서 접선각

$$s(s+3)(s^2+2s+2)+K=0$$

$$\angle G(s_1)H(s_1)=-(\theta_1+\theta_2+\theta_3+\theta_4)=(2i+1)180^\circ$$

$$-(135^\circ+\theta_2+90^\circ+26.6^\circ)=(2i+1)180^\circ$$



CHANGWON NATIONAL UNIVERSITY

근궤적의 성질

✓근궤적의 출발각과 도착각

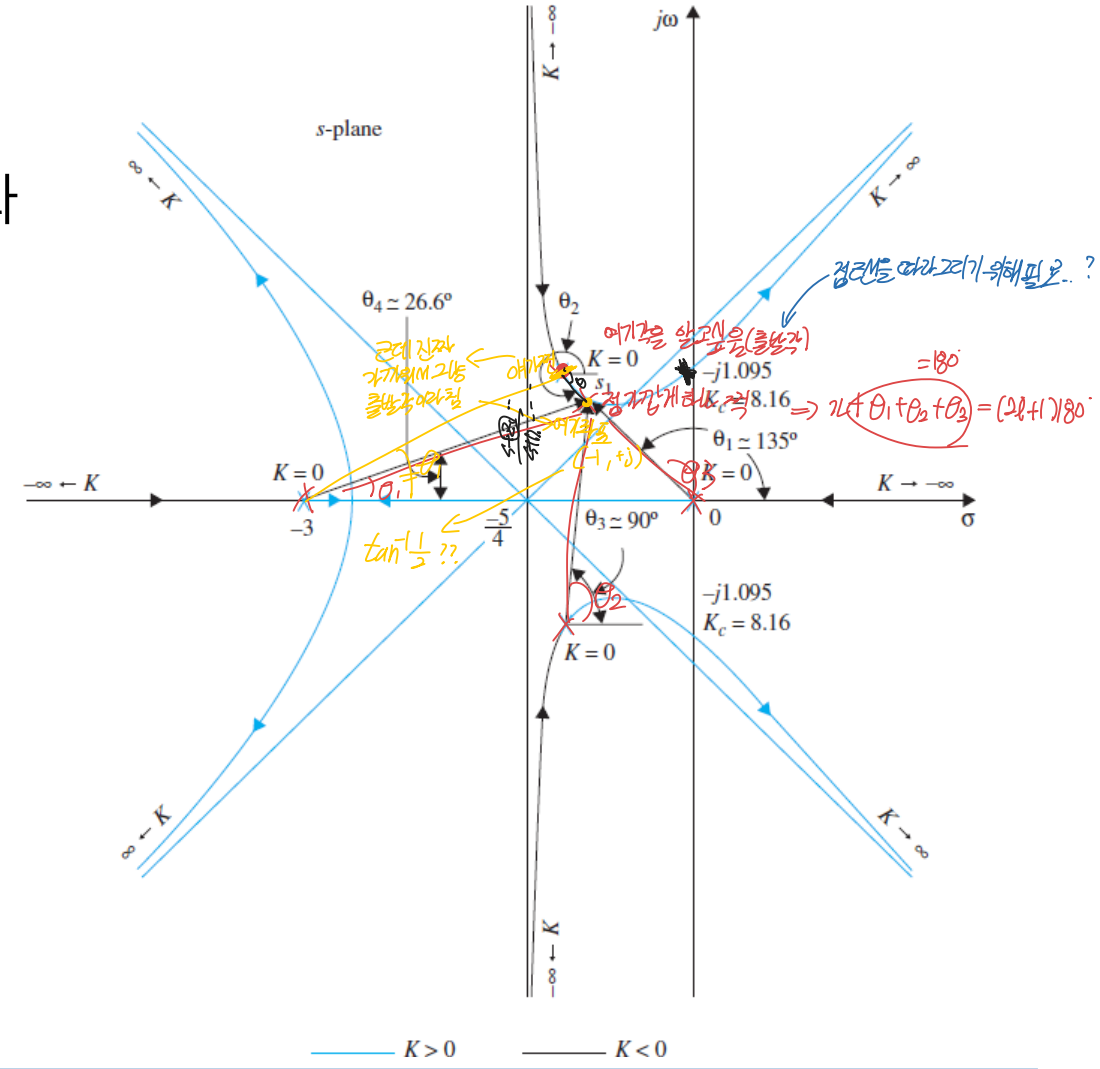
- G(s)H(s)의 극이나 영점에서 근궤적의 출발각과 도착각은 각각 그 점 부근의 궤적에서 접선각

$$s(s+3)(s^2+2s+2)+K=0$$

$$\angle G(s_1)H(s_1)=-(\theta_1+\theta_2+\theta_3+\theta_4)=(2i+1)180^\circ$$

$$-(135^\circ+\theta_2+90^\circ+26.6^\circ)=(2i+1)180^\circ$$

$S=0,-3,-1\pm j$



CHANGWON NATIONAL UNIVERSITY

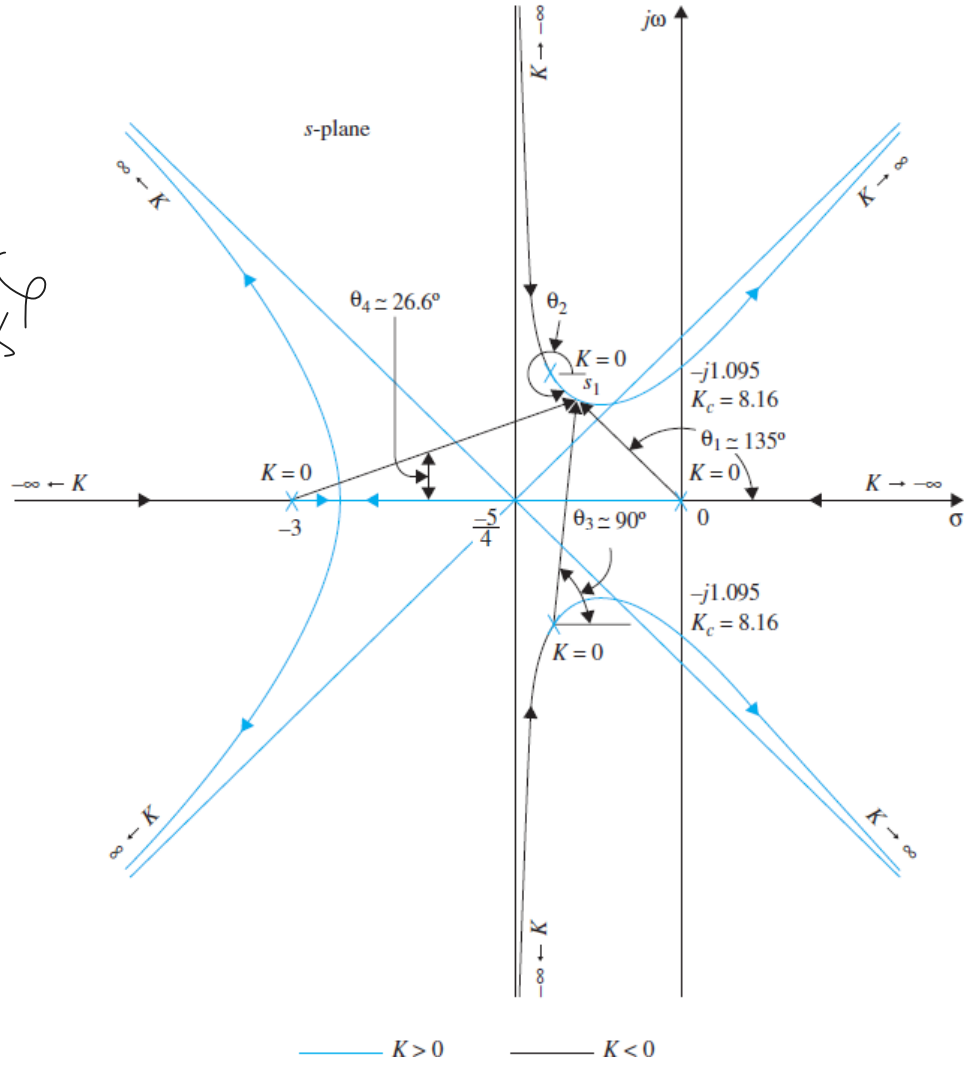
근궤적의 성질

✓근궤적과 허수축과의 교차

$$s(s+3)(s^2+2s+2)+K=0$$

s^4	1	8	K
s^3	5	6	0
s^2	$\frac{-\begin{vmatrix}1 & 8\\ 5 & 6\end{vmatrix}}{5}$	$\frac{-\begin{vmatrix}1 & K\\ 5 & 0\end{vmatrix}}{5}$	$\frac{-\begin{vmatrix}1 & 0\\ 5 & 0\end{vmatrix}}{5}$
	$=6.8$	$=K$	$=0$
s^1	$\frac{-\begin{vmatrix}5 & 6\\ 6.8 & K\end{vmatrix}}{6.8}$	$\frac{-\begin{vmatrix}5 & 0\\ 6.8 & 0\end{vmatrix}}{6.8}$	0
	$=\frac{-(5K-40.8)}{6.8}$	$=0$	
s^0	K	0	0

$\therefore K \le 8.16$



CHANGWON NATIONAL UNIVERSITY

근궤적의 성질

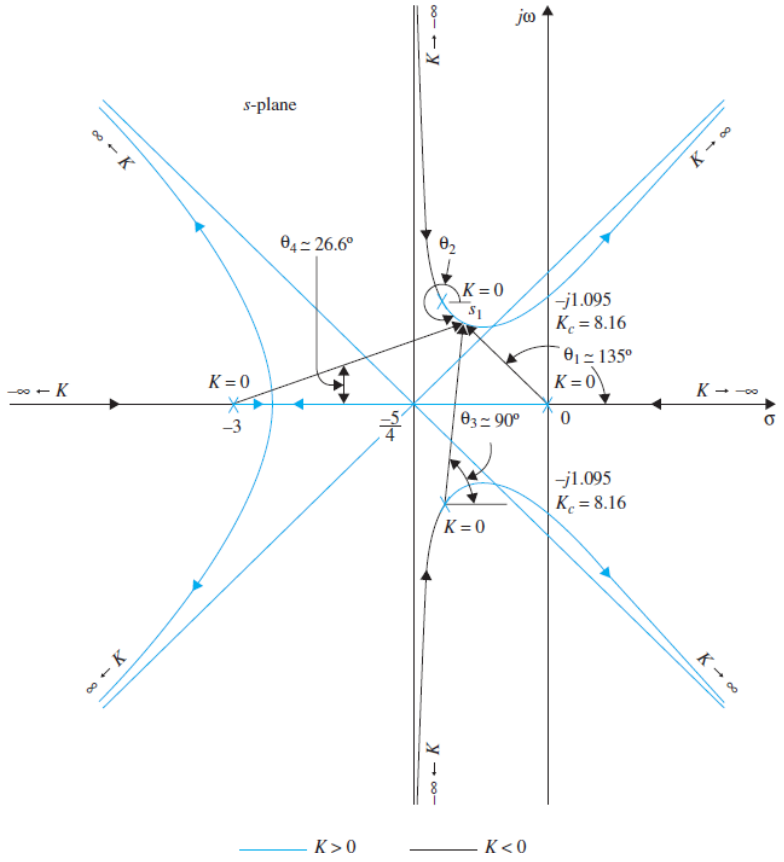
✓근궤적 상 이탈점 Breakaway points on the root roci

$$s(s+3)(s^2+2s+2)+K=0$$

$$\frac{dG_1(s)H_1(s)}{ds}=0 \qquad G_1(s)H_1(s)=-\frac{1}{K}$$

$$\frac{dK}{ds}=\frac{dG_1(s)H_1(s)/ds}{[G_1(s)H_1(s)]^2}$$

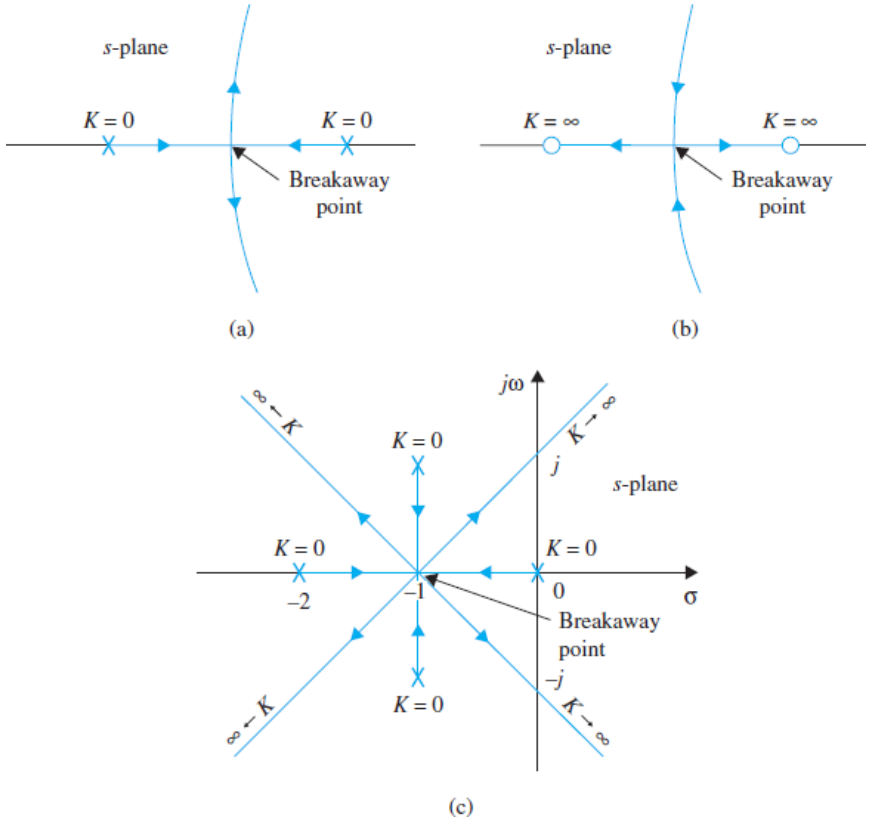
$$\frac{dK}{ds}=0$$



근궤적의 성질

✓이탈점에서 근궤적의 도착각과 출발각

✓근궤적이 이탈점에 도착하거나 출발하는 각은 그 점과 관계된 근궤적의 수에 따라 결정



근궤적의 성질

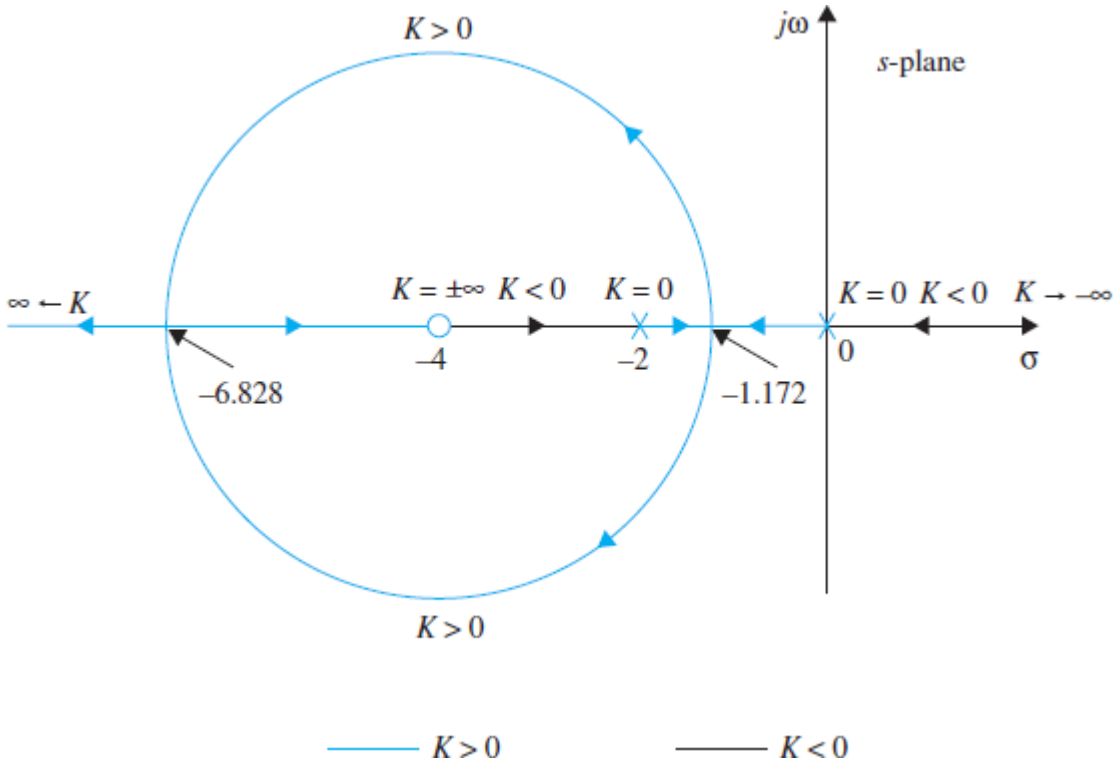
✓이탈점에서 근궤적의 도착각과 출발각

- 근궤적이 이탈점에 도착하거나 출발하는 각은 그 점과 관계된 근궤적의 수에 따라 결정

$$s(s+2)+K(s+4)=0$$

$$\frac{dG_1(s)H_1(s)}{ds}=\frac{s(s+2)-2(s+1)(s+4)}{s^2(s+2)^2}=0$$

$$s^2+8s+8=0$$



CHANGWON NATIONAL UNIVERSITY

근궤적의 성질

✓이탈점에서 근궤적의 도착각과 출발각

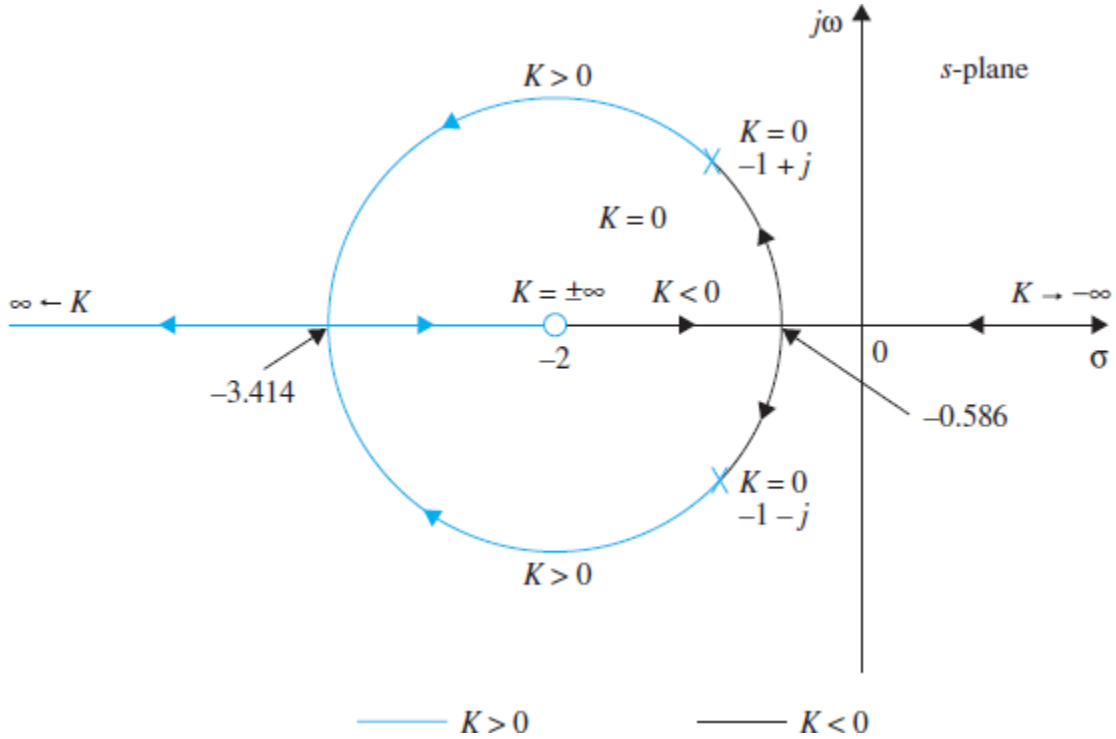
- 근궤적이 이탈점에 도착하거나 출발하는 각은 그 점과 관계된 근궤적의 수에 따라 결정

$$s^2+2s+2+K(s+2)=0$$

$$G(s)H(s)=\frac{K(s+2)}{s^2+2s+2}$$

$$\frac{dG_1(s)H_1(s)}{ds}=\frac{d}{ds}\left(\frac{s+2}{s^2+2s+2}\right)=\frac{s^2+2s+2-2(s+1)(s+2)}{(s^2+2s+2)^2}=0$$

$$s^2+4s+2=0$$



CHANGWON NATIONAL UNIVERSITY

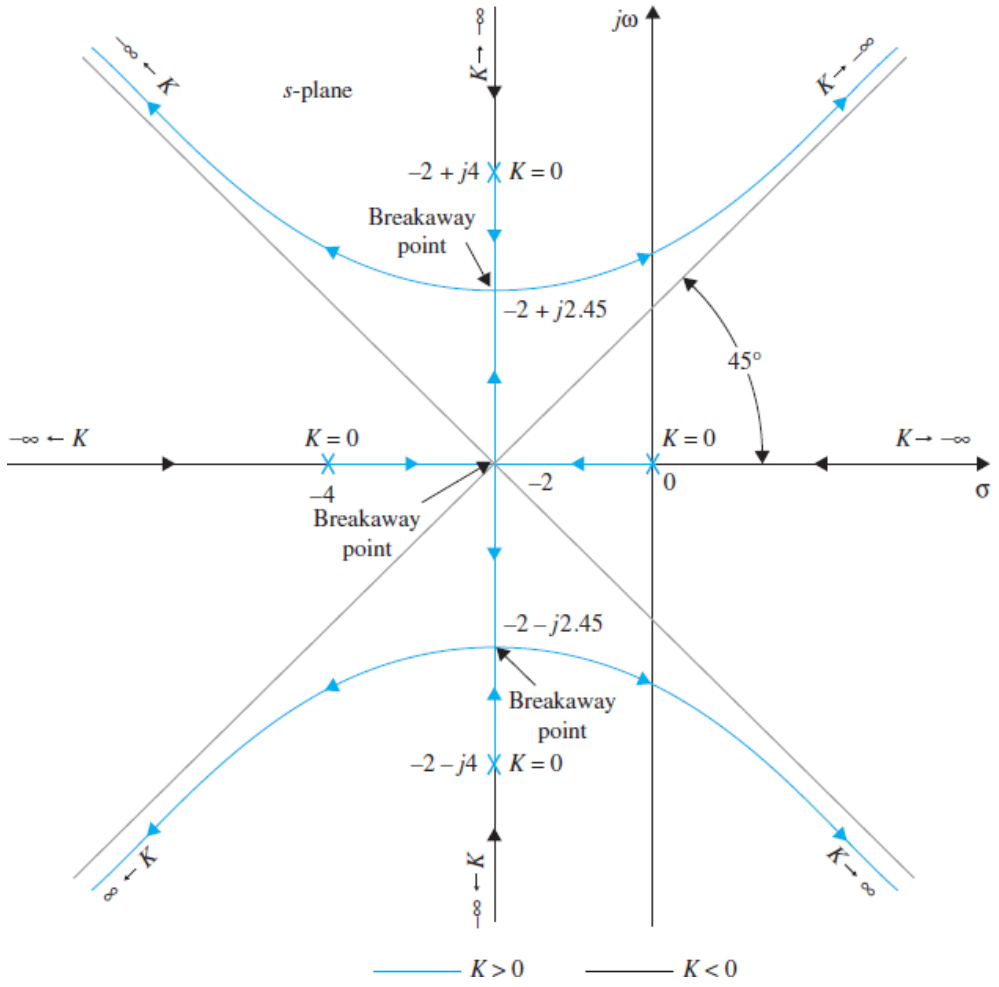
근궤적의 성질

$$s(s+4)(s^2+4s+20)+K=0$$

$$1+KG_1(s)H_1(s)=1+\frac{K}{s(s+4)(s^2+4s+20)}=0$$

$$\frac{dG_1(s)H_1(s)}{ds}=-\frac{4s^3+24s^2+72s+80}{[s(s+4)(s^2+4s+20)]^2}=0$$

$$s^3+6s^2+18s+20=0$$

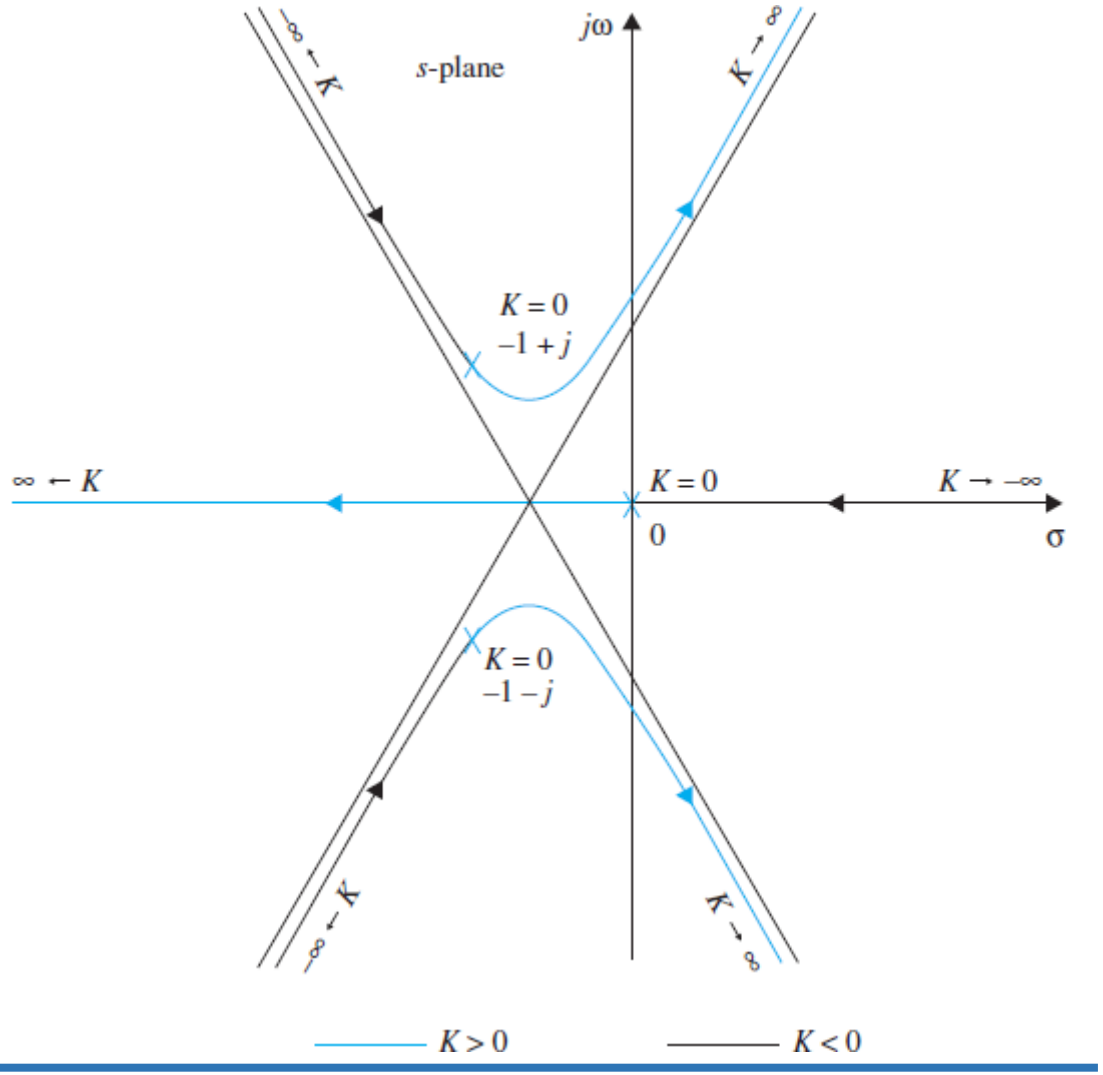


근궤적의 성질

$$s(s^2+2s+2)+K=0$$

$$1+KG_1(s)H_1(s)=1+\frac{K}{s(s^2+2s+2)}=0$$

$$3s^2+4s+2=0$$



CHANGWON NATIONAL UNIVERSITY

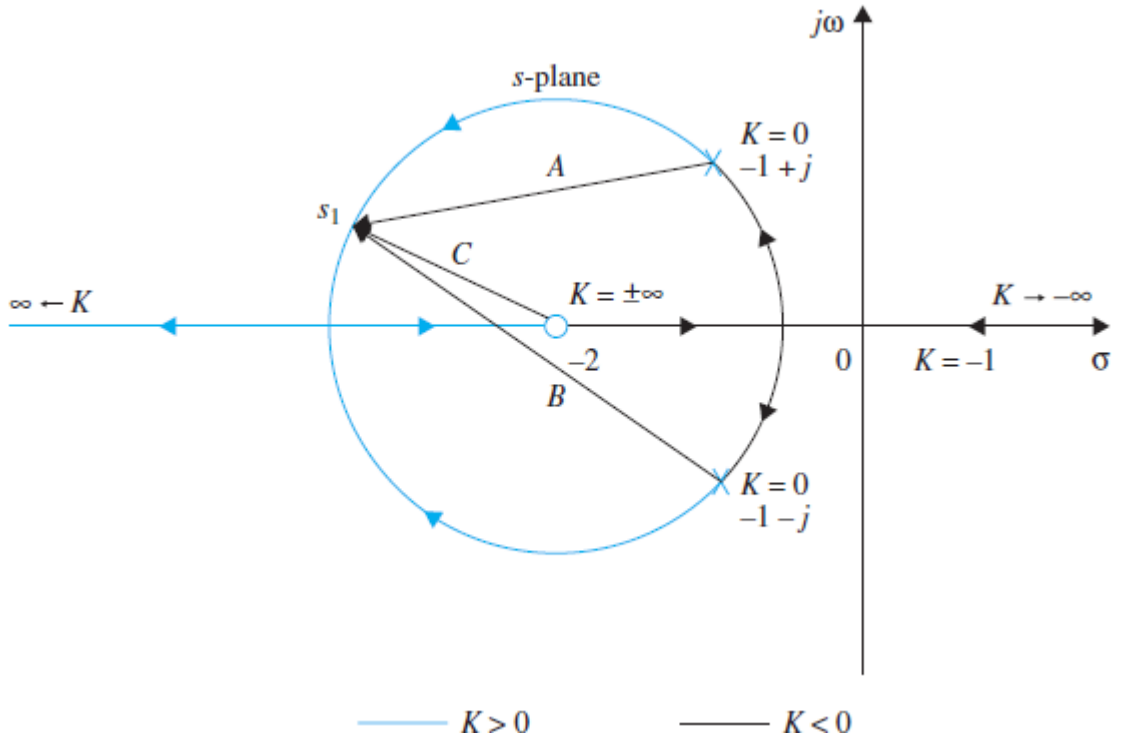
근궤적의 성질

✓근궤적에서의 K값 계산

$$|K| = \frac{\Pi \text{ lengths of vectors drawn from the poles of } G_1(s)H_1(s) \text{ to } s_1}{\Pi \text{ lengths of vectors drawn from the zeros of } G_1(s)H_1(s) \text{ to } s_1}$$

$$s^2 + 2s + 2 + K(s+2) = 0$$

$$K = \frac{A \times B}{C}$$



CHANGWON NATIONAL UNIVERSITY

근궤적의 성질

✓Summary

Table 9-1 Properties of the Root Loci of $1 + KG_1(s)H_1 = 0$

1. $K = 0$ points	The $K = 0$ points are at the poles of $G(s)H(s)$, including those at $s = \infty$.
2. $K = \pm\infty$ points	The $K = \infty$ points are at the zeros of $G(s)H(s)$, including those at $s = \infty$.
3. Number of separate root loci	The total number of root loci is equal to the order of the equation $1 + KG_1(s)H_1(s) = 0$.
4. Symmetry of root loci	The root loci are symmetrical about the axes of symmetry the of pole-zero configuration of $G(s)H(s)$.
5. Asymptotes of root loci as $s \to \infty$	For large values of s, the RL ($K > 0$) are asymptotic to asymptotes with angles given by $$\theta_i = \frac{2i+1}{\lvert n-m \rvert} \times 180^\circ$$ $\frac{2k+1}{P_{개수}-Z_{개수}}\pi \ (k=0,1,2,\cdots P-Z-1)$ For $K < 0$, the RL are asymptotic to $$\theta_i = \frac{2i}{\lvert n-m \rvert} \times 180^\circ$$ where $i = 0, 1, 2, \ldots, \lvert n-m \rvert - 1$, n = number of finite poles of $G(s)H(s)$, and m = number of finite zeros of $G(s)H(s)$.
6. Intersection of the asymptotes	(a) The intersection of the asymptotes lies only on the real axis in the s-plane. (b) The point of intersection of the asymptotes is given by $$\sigma_1 = \frac{\sum \text{real parts of poles of } G(s)H(s) - \sum \text{real parts of zeros of } G(s)H(s)}{n-m}$$
7. Root loci on the real axis	RL for $K \geq 0$ are found in a section of the real axis only if the total number of real poles and zeros of $G(s)H(s)$ to the **right** of the section is **odd**. If the total number of real poles and zeros to the right of a given section is **even**, RL for $K \leq 0$ are found.
8. Angles of departure	The angle of departure or arrival of the RL from a pole or a zero of $G(s)H(s)$ can be determined by assuming a point s_1 that is very close to the pole, or zero, and applying the equation $$\angle G(s_1)H(s_1) = \sum_{k=1}^{m} \angle(s_1 - z_k) - \sum_{j=1}^{n} \angle(s_1 - p_j)$$ $$= 2(i+1)180^\circ \quad K \geq 0$$ $$= 2i \times 180^\circ \quad K \leq 0$$ where $i = 0, \pm 1, \pm 2, \ldots$.
9. Intersection of the root loci with the imaginary axis	The crossing points of the root loci on the imaginary axis and the corresponding values of K may be found by use of the Routh-Hurwitz criterion.
10. Breakaway points	The breakaway points on the root loci are determined by finding the roots of $dK/ds = 0$, or $dG(s)H(s)/ds = 0$. These are necessary conditions only.
11. Calculation of the values of K	The absolute value of K at any point s_1 on the root loci is on the root loci determined from the equation $$\lvert K \rvert = \frac{1}{\lvert G_1(s_1)H_1(s_1) \rvert}$$

근궤적의 성질

$$s(s+5)(s+6)(s^2+2s+2)+K(s+3)=0$$

$$G(s)H(s)=\frac{K(s+3)}{s(s+5)(s+6)(s^2+2s+2)}$$

1. The $K = 0$ points are at the poles of $G(s)H(s)$: $s = -5, -6, -1 + j$, and $-1 - j$.

2. The $K = \pm\infty$ points are at the zeros of $G(s)H(s)$: $s = -3, \infty\cdot, \infty, \infty$.

3. There are five separate branches on the root loci.

4. The root loci are symmetrical with respect to the real axis of the s-plane.

5. Since $G(s)H(s)$ has five poles and one finite zero, four RL and CRL should approach infinity along the asymptotes. The angles of the asymptotes of the RL are given by

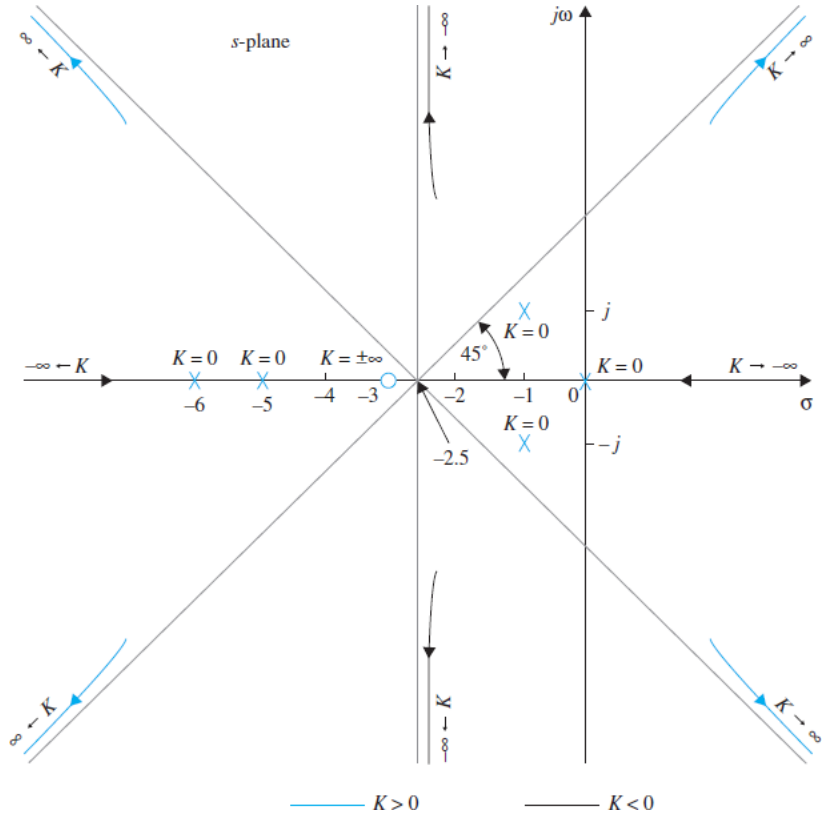
$$\theta_i=\frac{2i+1}{|n-m|}180^\circ=\frac{2i+1}{|5-1|}180^\circ \quad 0\le K<\infty \qquad (9\text{-}67)$$

6. The intersection of the asymptotes is given by

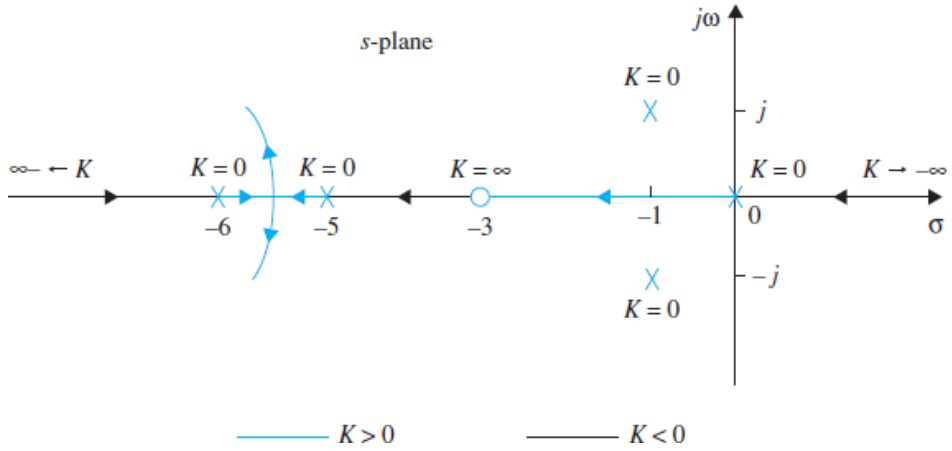
$$\sigma_1=\frac{\Sigma(-5-6-1-1)-(-3)}{4}=-2.5$$

근궤적의 성질

$$s(s+5)(s+6)(s^2+2s+2)+K(s+3)=0 \qquad G(s)H(s)=\frac{K(s+3)}{s(s+5)(s+6)(s^2+2s+2)}$$



7. Root loci on the real axis



CHANGWON NATIONAL UNIVERSITY

근궤적의 성질

$$s(s+5)(s+6)(s^2+2s+2)+K(s+3)=0 \qquad G(s)H(s)=\frac{K(s+3)}{s(s+5)(s+6)(s^2+2s+2)}$$

8. Angles of departure:

$$\angle(s_1+3)-\angle s_1-\angle(s_1+1+j)-\angle(s_1+5)-\angle(s_1+1-j)=(2i+1)180^\circ$$

$$26.6^\circ-135^\circ-90^\circ-14^\circ-11.4^\circ-\theta\cong(2i+1)180^\circ$$

$$\theta'=180^\circ-43.8^\circ=136.2^\circ$$

9. The intersection of the root loci on the imaginary axis is determined using Routh's tabulation.

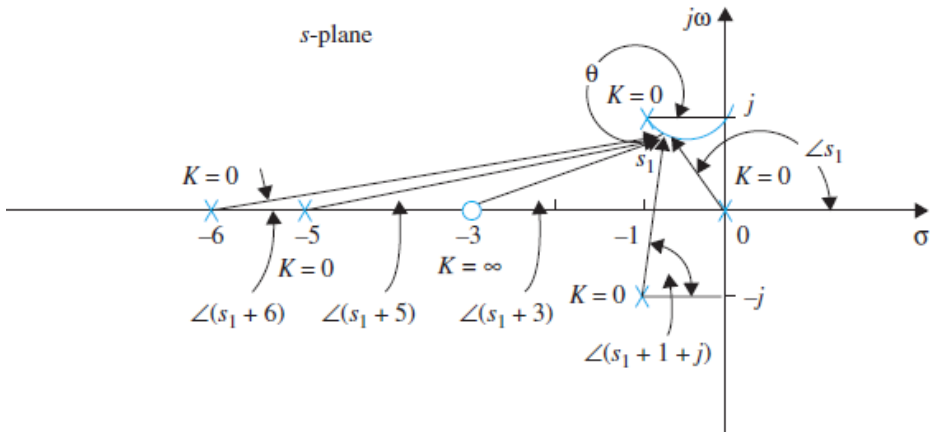
$$s^5+13s^4+54s^3+82s^2+(60+K)s+3K=0$$

$$65.6-0.212K>0 \quad \text{or} \quad K<309$$

$$3940-105K-0.163K^2>0 \quad \text{or} \quad K<35$$

$$K>0$$

$K = 35$ and $K = 0$.



CHANGWON NATIONAL UNIVERSITY

근궤적의 성질

$$s(s+5)(s+6)(s^2+2s+2)+K(s+3)=0 \qquad G(s)H(s)=\frac{K(s+3)}{s(s+5)(s+6)(s^2+2s+2)}$$

10. Breakaway points

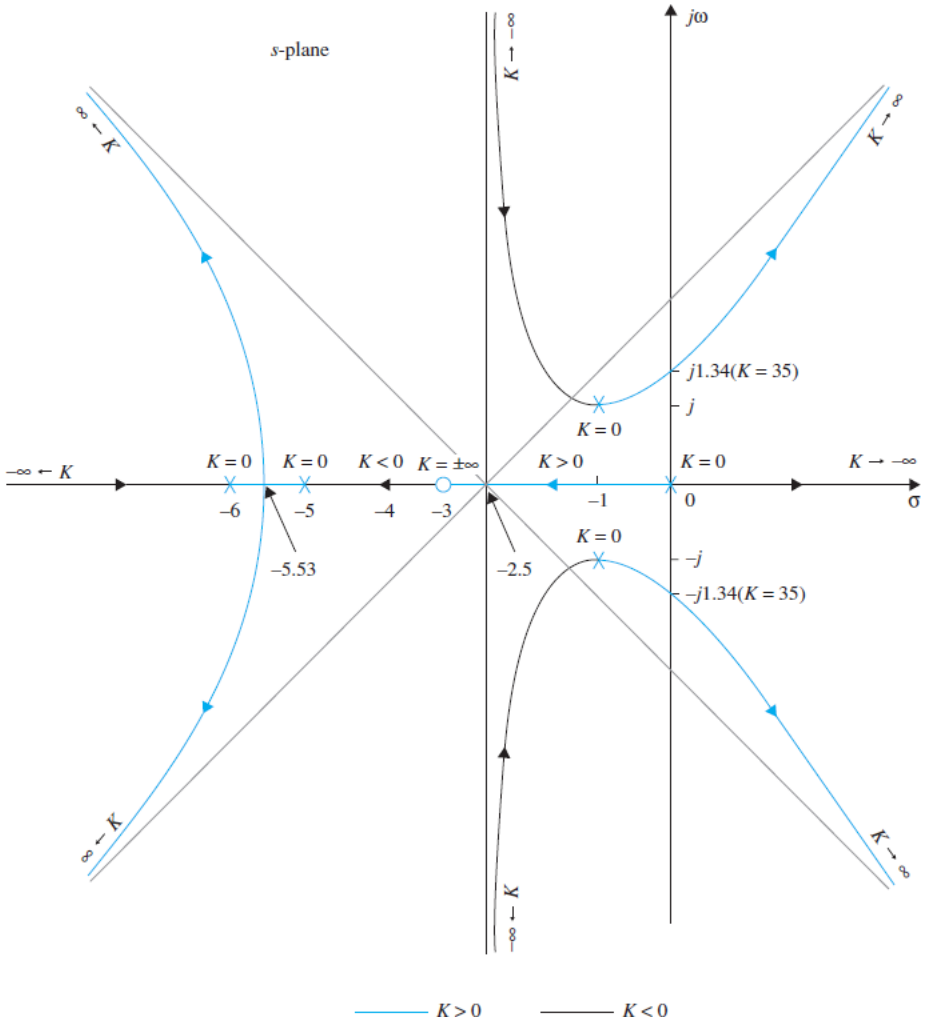
$$s^5+13.5s^4+66s^3+142s^2+123s+45=0$$

K를 증가시키면...?

$s=3.33+j1.204 \quad s=3.33-j1.204$

$s=-0.656+j0.468 \quad s=-0.656-j0.468$

$s=-5.53$



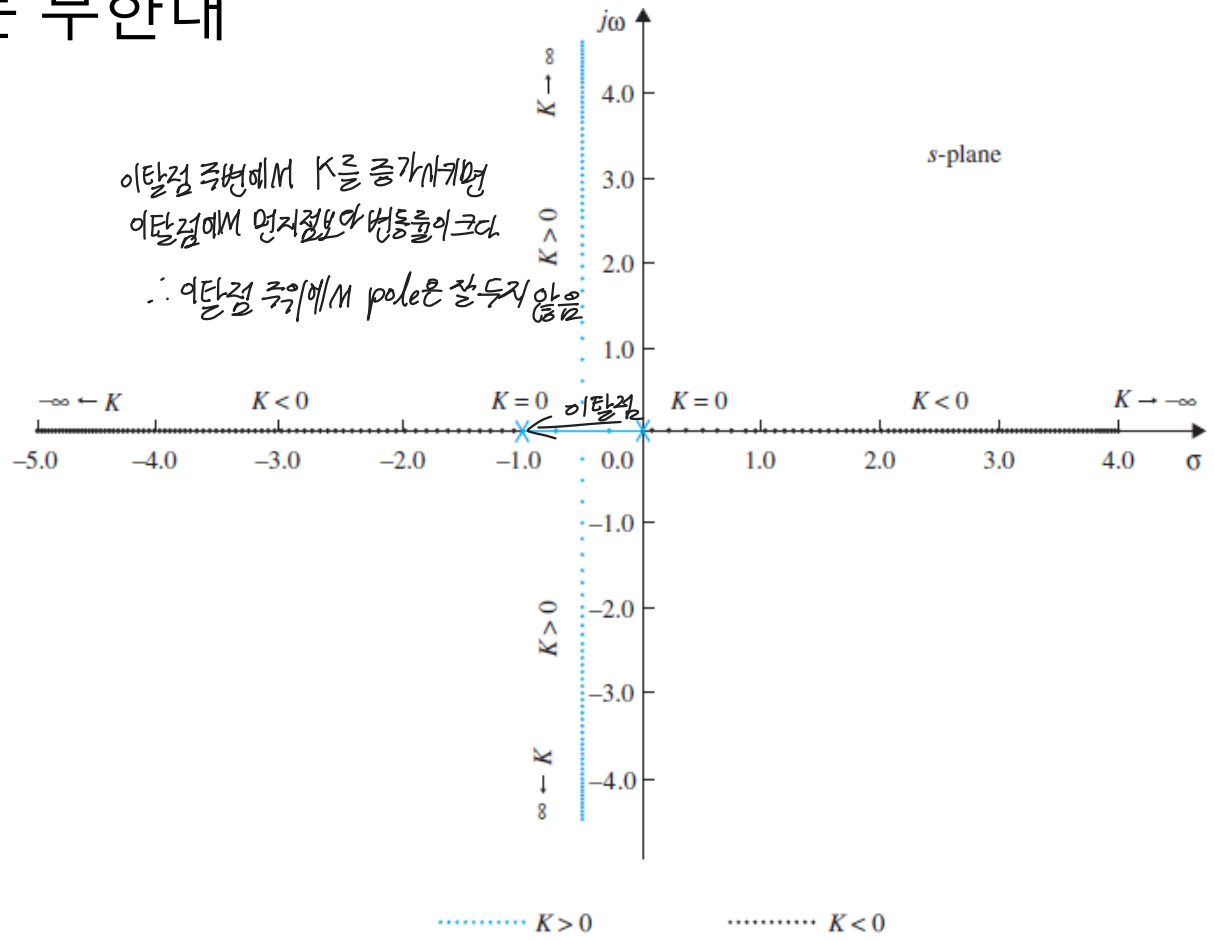
CHANGWON NATIONAL UNIVERSITY

근의 민감도 The root sensitivity

✓이탈점에서의 근의 민감도는 무한대

$$S_K = \frac{ds/s}{dK/K} = \frac{K}{s}\frac{ds}{dK}$$

$$s(s+1)+K=0$$



CHANGWON NATIONAL UNIVERSITY

근궤적 작도에 있어서 중요한 점

✓G(s)H(s)에 극 추가

- G(s)H(s)에 극을 추가하는 것은 근궤적을 우반면 쪽으로 이동시키는 효과

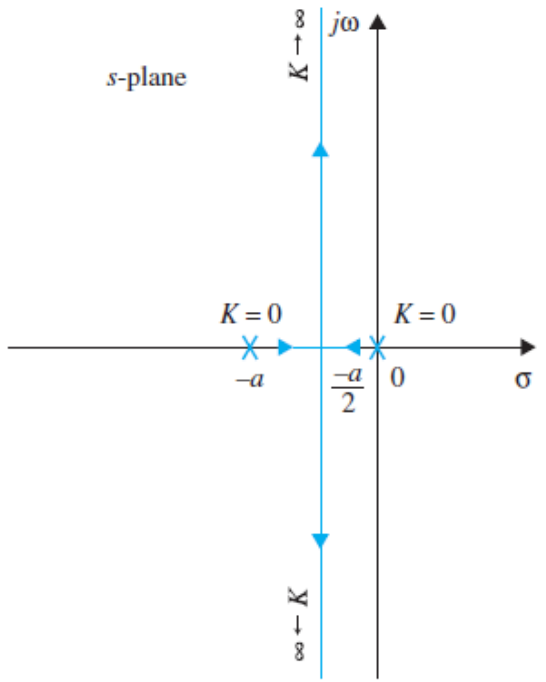
$$G(s)H(s)=\frac{K}{s(s+a)} \qquad a>0$$

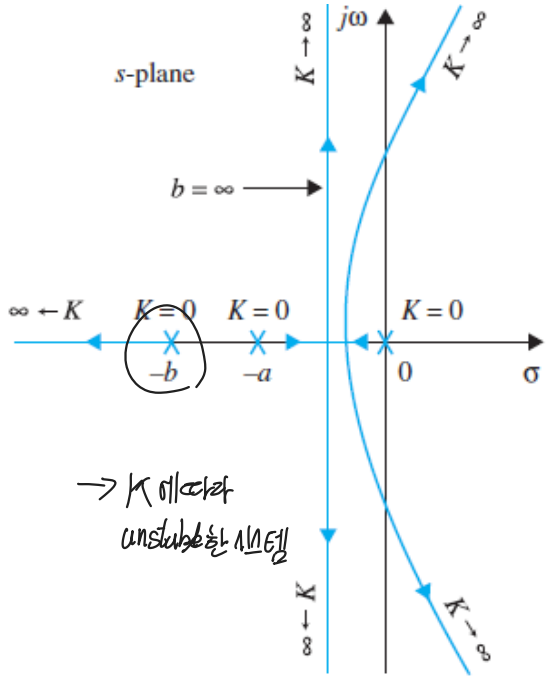
$$G(s)H(s)=\frac{K}{s(s+a)(s+b)}$$

$n=0$

$n=2\rightarrow 3$

$-a/2$ to $-(a+b)/2$





→ K에 따라 unstable한 시스템

CHANGWON NATIONAL UNIVERSITY

근궤적 작도에 있어서 중요한 점

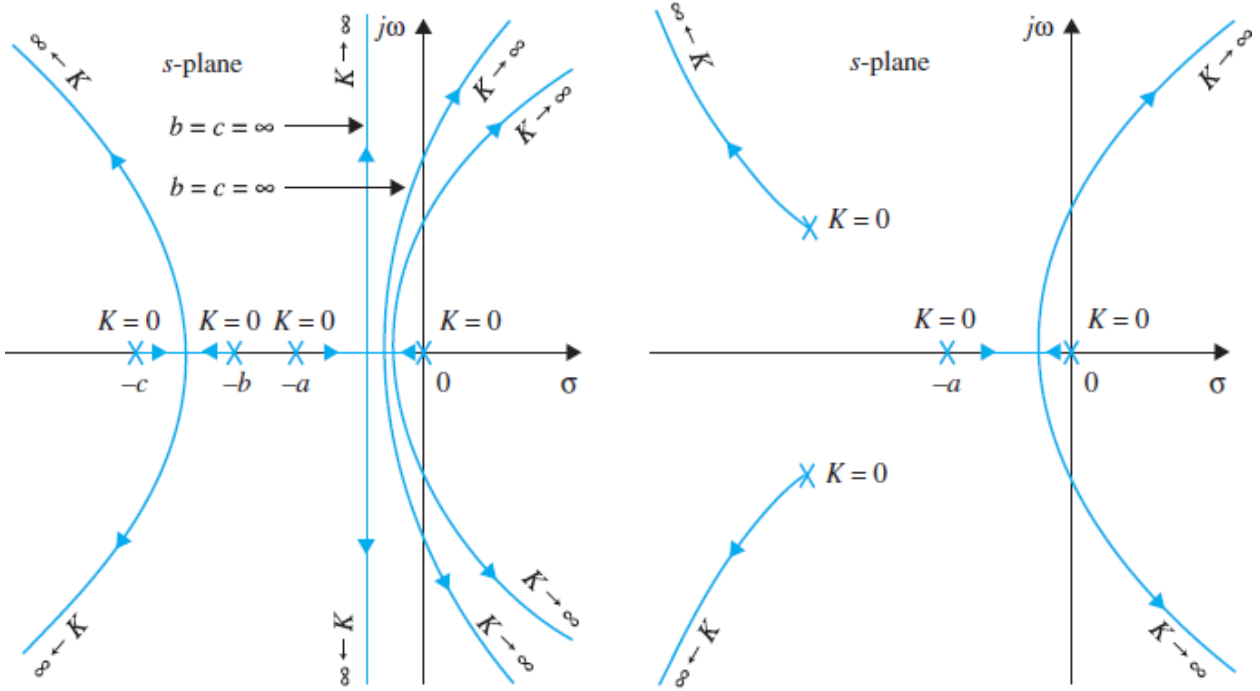
✓G(s)H(s)에 극 추가

- G(s)H(s)에 극을 추가하는 것은 근궤적을 우반면 쪽으로 이동시키는 효과

$$G(s)H(s)=\frac{K}{s(s+a)} \qquad a>0$$

$$G(s)H(s)=\frac{K}{s(s+a)(s+b)}$$

$$s=-c,\ c>b$$



CHANGWON NATIONAL UNIVERSITY

근궤적 작도에 있어서 중요한 점

✓G(s)H(s)에 영점 추가

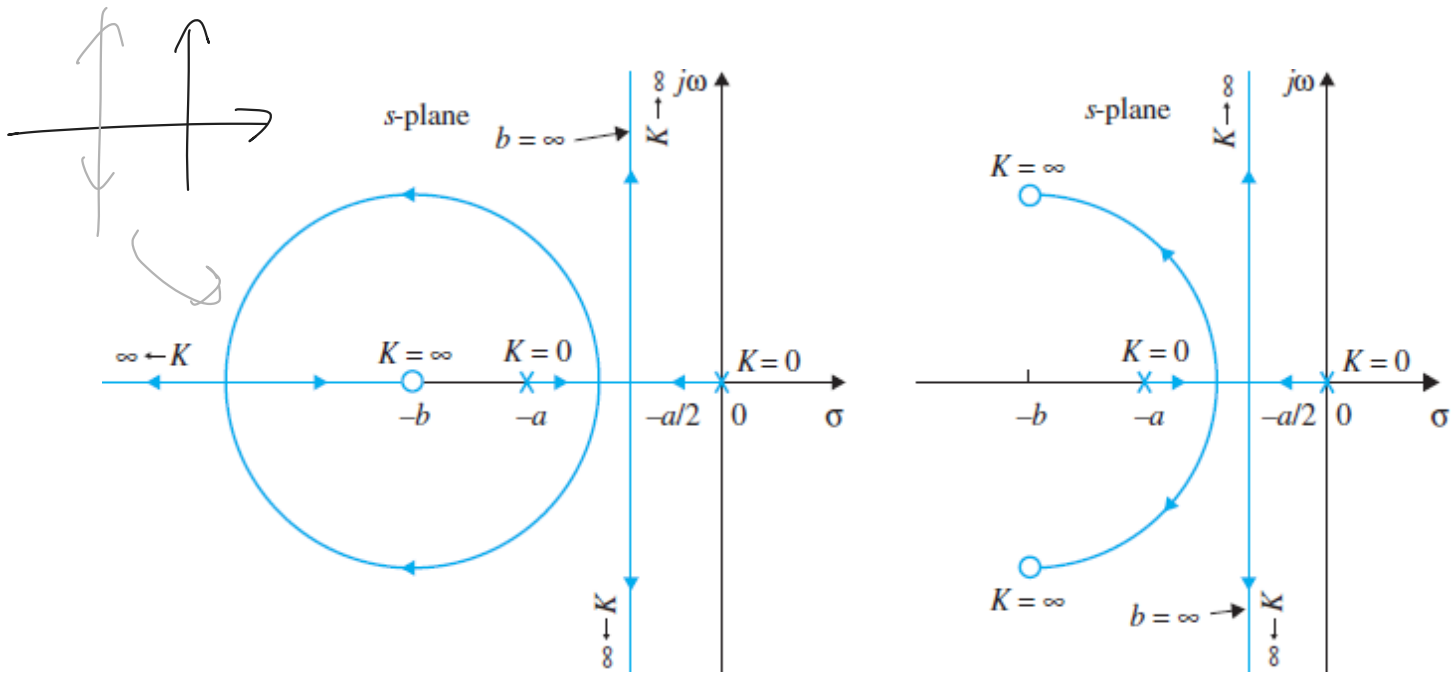
- G(s)H(s)에 좌반면 영점을 추가하면 일반적으로 근궤적 s평면 좌반면 쪽으로 구부리고 이동시키는 효과

$$G(s)H(s)=\frac{K}{s(s+a)} \qquad a>0$$

(s+b)추가

$$\theta=\frac{(2i+1)}{n-m}180^\circ$$

90° 270° → 180°



CHANGWON NATIONAL UNIVERSITY

근궤적: 다중 파라미터의 변화

- 두 개의 가변 파라미터를 가지는 방정식의 근궤적

$$s^3 + K_2 s^2 + K_1 s + K_1 = 0$$

$$s^3 + K_1 s + K_1 = 0 \qquad K_2 = 0$$

$$1 + \frac{K_1(s+1)}{s^3} = 0$$

$$G_1(s)H_1(s) = \frac{s+1}{s^3}$$

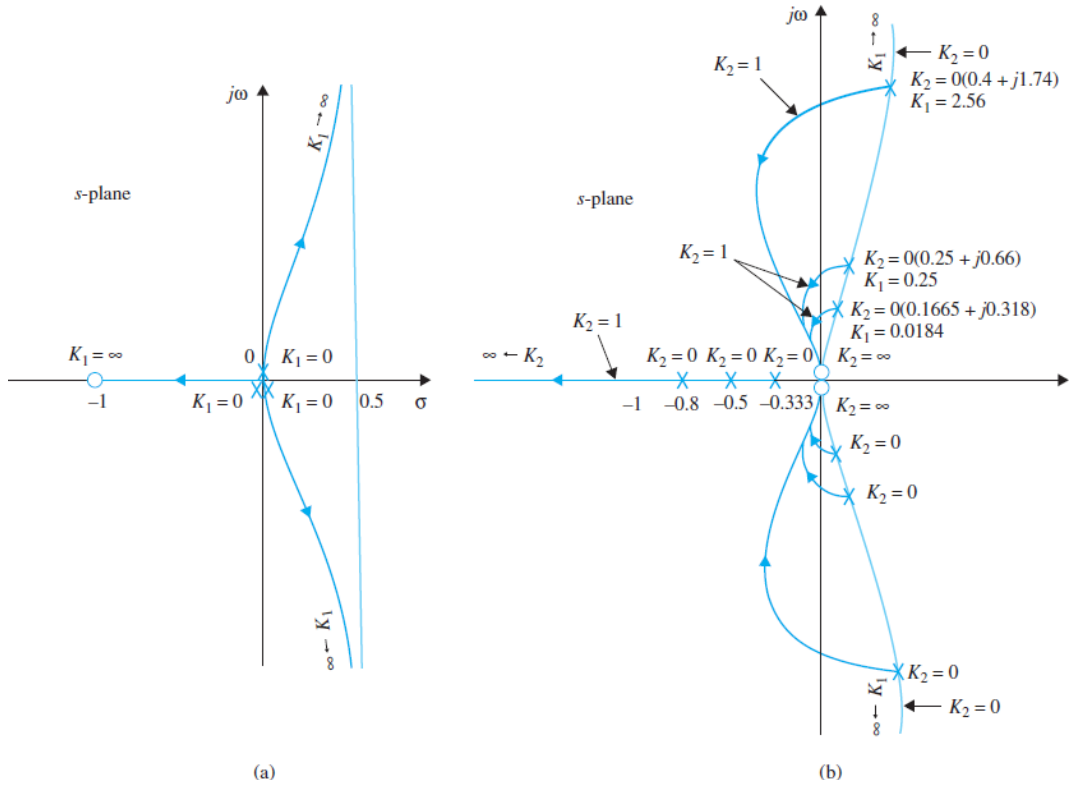


Figure 9-27 Root contours of $s^2 + K_2 s^2 + K_1 s + K_1 = 0$. (*a*) $K_2 = 0$. (*b*) K_2 varies and K_1 is a constant.

근컨투어: 다중 파라미터의 변화

✓두 개의 가변 파라미터를 가지는 방정식의 근궤적

$$s^3 + K_2 s^2 + K_1 s + K_1 = 0$$

$$s^3 + K_1 s + K_1 = 0 \qquad K_2 = 0$$

$$1 + \frac{K_1(s+1)}{s^3} = 0$$

$$G_1(s)H_1(s) = \frac{s+1}{s^3}$$

